Mathematical Analysis III

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Introduction

The third volume of these notes continues the development of mathematical analysis beyond the differential and integral calculus of single- and multivariable functions. The main objects are no longer individual functions alone, but also vector fields, infinite series of functions, improper integrals, Fourier expansions, and integrals depending on parameters. The first two volumes establish the basic language of analysis: real numbers, limits, mappings, derivatives, differentials, integrals, multivariable calculus, multiple integrals, line and surface integrals, and the elementary topology of Euclidean space. This volume develops several directions that naturally arise from those foundations.

Chapter 1 studies the mathematics of fields. Gradient, divergence, and curl describe three different aspects of local variation: the direction of maximal increase of a scalar field, the local flux of a vector field, and its local circulation. Green's formula, Gauss's formula, and Stokes's formula connect these differential quantities with global integral quantities. Potential fields, conservative fields, solenoidal fields, and vector potentials then give a more structural description of vector calculus.

Chapter 2 returns to infinite processes through numerical series. The basic problem is to determine when an infinite sum

$$\sum_{n=1}^{\infty} a_n$$

has a well-defined value. Comparison tests, the Cauchy root test, D'Alembert's ratio test, Raabe's test, Gauss's test, Dirichlet's test, and Abel's test provide different ways of controlling the behavior of the tails of a series. Absolute convergence and conditional convergence reveal that convergence may also depend on the order in which infinitely many terms are summed.

Chapter 3 replaces numerical terms by functions. This leads to the distinction between pointwise convergence and uniform convergence. Uniform convergence is especially important because it allows continuity, integration, and differentiation to pass through the limit under suitable assumptions. Power series form the most important class of function series in this chapter and connect infinite series directly with Taylor expansions.

Chapter 4 develops improper integrals. Infinite intervals and singularities force the ordinary definition of the definite integral to be replaced by a limiting process. The Cauchy, Dirichlet, and Abel criteria again appear, now in integral form. The chapter also introduces principal values and improper multiple integrals, extending the same ideas to higher dimensions.

Chapter 5 is devoted to Fourier analysis. A function is represented through trigonometric oscillations of different frequencies,

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin nx\right).$$

The central questions are how the Fourier coefficients are determined, when the Fourier series converges to the original function, and in what sense the approximation is optimal. Dirichlet's theorem, Fejer's theorem, Bessel's inequality, Parseval's identity, and the Fourier integral provide several different answers to these questions.

The final chapter studies integrals depending on a parameter. Expressions such as

$$\Phi(u) = \int_a^{+\infty} f(x,u)\,dx$$

contain both integration and limiting processes. The main issue is therefore when these operations may be interchanged with limits, differentiation, or another integration. Uniform convergence again becomes the key condition. The chapter ends with the Gamma and Beta functions, which provide classical examples of parameter-dependent improper integrals and connect analysis with factorials, special functions, and many important definite integrals.

Taken together, these six chapters show a common theme of advanced calculus: local operations, infinite limiting processes, and global integral structures are closely related. Many of the results in this volume are not isolated techniques, but different expressions of the same principle—to understand when an operation remains valid after passing to a limit.

Chapter 1

Mathematics of Fields

Before introducing the differential operators of vector analysis, we recall the three integral formulas used throughout this chapter.

Green's formula. Let $D \subset \mathbb{R}^2$ be a plane region with positively oriented boundary ∂D. Then

$$\oint_{\partial D} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.$$

Gauss's formula. Let $V \subset \mathbb{R}^3$ be a spatial region with positively oriented boundary surface ∂V. Then

$$\begin{aligned}
&\iint_{\partial V} P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy \\
&\quad = \iint_{\partial V} (P\cos\alpha + Q\cos\beta + R\cos\gamma)\,d\sigma \\
&\quad = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz,
\end{aligned}$$

where

$$\mathbf{n} = (\cos\alpha, \cos\beta, \cos\gamma)$$

is the outward unit normal vector.

Stokes's formula. Let Σ be an oriented surface with compatible oriented boundary $\partial\Sigma$. Then

$$\begin{aligned}
&\oint_{\partial\Sigma} P\,dx + Q\,dy + R\,dz \\
&\quad = \iint_\Sigma \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) dy\,dz \\
&\qquad + \iint_\Sigma \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right) dz\,dx \\
&\qquad + \iint_\Sigma \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.
\end{aligned}$$

Equivalently,

$$\oint_{\partial\Sigma} P\,dx + Q\,dy + R\,dz = \iint_\Sigma \begin{vmatrix} \cos\alpha & \cos\beta & \cos\gamma \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} d\sigma.$$

1.1 Gradient of a Scalar Field

Let $D \subset \mathbb{R}^3$ be open, and let

$$f : D \to \mathbb{R}$$

be continuously differentiable.

Definition 1.1.1 (Gradient)**.** For $p \in D$, define

$$\operatorname{grad} f(p) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)_p.$$

The vector $\operatorname{grad} f(p)$ is called the *gradient* of f at p.

The gradient points in the direction of the maximal directional derivative.
Indeed, if $\mathbf{e}$ is a unit vector, then

$$D_{\mathbf{e}} f(p) = \nabla f(p) \cdot \mathbf{e}.$$

Therefore, by the Cauchy–Schwarz inequality,

$$D_{\mathbf{e}} f(p) \leq \|\nabla f(p)\|,$$

and equality holds when $\mathbf{e}$ has the same direction as $\nabla f(p)$.
For a constant c, the set

$$\{p \in D : f(p) = c\}$$

is called a *level surface* of f.

Proposition 1.1.2. At every regular point of a level surface $f = c$, the vector ∇f is normal to the level surface.

Proof. Let

$$\mathbf{r}(t) = \Big(x(t), y(t), z(t)\Big)$$

be a differentiable curve contained in the level surface. Then

$$f(\mathbf{r}(t)) = c.$$

Differentiating gives

$$\nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

Thus ∇f is orthogonal to every tangent vector of the level surface. □

Introduce the *Nabla operator*

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right).$$

Then

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right).$$

The gradient satisfies

$$\nabla(cf) = c\nabla f,$$
$$\nabla(f \pm g) = \nabla f \pm \nabla g,$$

and

$$\nabla(fg) = f\nabla g + g\nabla f.$$

If φ is a differentiable function of one variable, then

$$\nabla(\varphi \circ f) = \varphi'(f)\nabla f.$$

1.2 Divergence of a Vector Field

Let

$$F : D \subset \mathbb{R}^3 \to \mathbb{R}^3$$

be a vector field, and let Σ be an oriented surface in D with unit normal vector $\mathbf{n}$.

Definition 1.2.1 (Flux)**.** The integral

$$\iint_\Sigma F \cdot \mathbf{n}\, d\sigma$$

is called the *flux* of F through Σ.

The orientation is taken to be positive when the normal points outward and negative when the normal points inward.
Let S be a closed surface surrounding a point M, and let V be the spatial region enclosed by S. Denote its volume by $m(V)$.

Definition 1.2.2 (Divergence)**.** The divergence of F at M is defined by

$$(\operatorname{div} F)_M = \lim_{V \to M} \frac{1}{m(V)} \iint_S F \cdot \mathbf{n}\, d\sigma,$$

provided that the limit exists.

Thus divergence measures the flux per unit volume in the limit.
Let

$$F = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k},$$

where P, Q, R have continuous first partial derivatives.

Theorem 1.2.3. The divergence of F is

$$\operatorname{div} F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Sketch of Proof. Let

$$\mathbf{n} = (\cos\alpha, \cos\beta, \cos\gamma).$$

Then

$$F \cdot \mathbf{n} = P\cos\alpha + Q\cos\beta + R\cos\gamma.$$

By Gauss's formula,

$$\iint_S F \cdot \mathbf{n}\, d\sigma = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dV.$$

By the mean value theorem for multiple integrals,

$$\frac{1}{m(V)} \iint_S F \cdot \mathbf{n}\, d\sigma = \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right)\Big|_{\xi_V}$$

for some $\xi_V \in V$. Letting $V \to M$ gives the result. □

Using the Nabla operator,

$$\operatorname{div} F = \nabla \cdot F.$$

The divergence satisfies

$$\nabla \cdot (cF) = c\nabla \cdot F,$$
$$\nabla \cdot (F_1 + F_2) = \nabla \cdot F_1 + \nabla \cdot F_2.$$

If φ is a scalar field, then

$$\nabla \cdot (\varphi F) = \varphi \nabla \cdot F + F \cdot \nabla\varphi.$$

1.3 Curl of a Vector Field

Let F be a vector field on $D \subset \mathbb{R}^3$, and let Γ be a closed curve in D. Let $\mathbf{t}$ denote the positively oriented unit tangent vector of Γ.

Definition 1.3.1 (Circulation)**.** The quantity

$$\oint_\Gamma F \cdot \mathbf{t}\, ds$$

is called the *circulation* of F along Γ.

The average circulation around a small closed curve is

$$\frac{1}{A} \oint_\Gamma F \cdot \mathbf{t}\, ds,$$

where A is the area enclosed by Γ.
The limiting value as the enclosed area shrinks to a point measures the local strength of rotation.
Let

$$F = P(x,y,z)\mathbf{i} + Q(x,y,z)\mathbf{j} + R(x,y,z)\mathbf{k}.$$

Definition 1.3.2 (Curl)**.** The *curl* of F is

$$\begin{aligned} \operatorname{rot} F = &\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} \\ &+ \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} \\ &+ \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}. \end{aligned}$$

Equivalently,

$$\operatorname{rot} F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

Using the Nabla operator,

$$\operatorname{rot} F = \nabla \times F.$$

Remark 1.3.3. The direction of $\operatorname{rot} F$ gives the direction in which the local circulation density is maximal, while its magnitude gives the maximal local circulation strength.

Indeed, by Stokes's formula,

$$\oint_\Gamma F \cdot \mathbf{t}\, ds = \iint_\Sigma (\operatorname{rot} F) \cdot \mathbf{n}\, d\sigma.$$

If Σ shrinks to M, then

$$\frac{1}{A} \oint_\Gamma F \cdot \mathbf{t}\, ds \to (\operatorname{rot} F)(M) \cdot \mathbf{n}.$$

This is maximal when $\mathbf{n}$ has the same direction as $(\operatorname{rot} F)(M)$.
The curl satisfies

$$\nabla \times (cF) = c\nabla \times F,$$

$$\nabla \times (F_1 + F_2) = \nabla \times F_1 + \nabla \times F_2.$$

For a scalar field φ,

$$\nabla \times (\varphi F) = \varphi \nabla \times F + \nabla \varphi \times F.$$

For two vector fields F_1, F_2,

$$\nabla \cdot (F_1 \times F_2) = (\nabla \times F_1) \cdot F_2 - (\nabla \times F_2) \cdot F_1.$$

1.4 Potential Fields and Potential Functions

Definition 1.4.1 (Potential Field)**.** Let

$$F = (P, Q, R)$$

be a vector field on $D \subset \mathbb{R}^3$. If there exists a scalar field φ on D such that

$$\nabla \varphi(p) = F(p)$$

for every $p \in D$, then F is called a *potential field*, and φ is called a *potential function* of F.

Thus

$$\frac{\partial \varphi}{\partial x} = P, \qquad \frac{\partial \varphi}{\partial y} = Q, \qquad \frac{\partial \varphi}{\partial z} = R.$$

If $A, B \in D$, then

$$\varphi(B) - \varphi(A) = \int_A^B F \cdot d\mathbf{p}.$$

Hence the potential function is unique up to an additive constant.

Definition 1.4.2 (Conservative Field)**.** A vector field F is called *conservative* if

$$\oint_\Gamma F \cdot d\mathbf{p} = 0$$

for every closed curve $\Gamma \subset D$.

Equivalently, if Γ_1 and Γ_2 have the same initial and terminal points, then

$$\int_{\Gamma_1} F \cdot d\mathbf{p} = \int_{\Gamma_2} F \cdot d\mathbf{p}.$$

Definition 1.4.3 (Irrotational Field)**.** A vector field F is called *irrotational* if

$$\operatorname{rot} F = \nabla \times F = 0$$

throughout its domain.

Definition 1.4.4 (Simply Connected Region)**.** A region $\Omega \subset \mathbb{R}^3$ is called *simply connected* if the interior of every closed curve in Ω lies entirely in Ω.

Theorem 1.4.5. Let $D \subset \mathbb{R}^3$ be simply connected and let $F \in C^2(D)$. Then the following statements are equivalent:

F is a potential field,

F is irrotational,

F is conservative.

Proof. Suppose first that F is a potential field. Then

$$F = \nabla\varphi.$$

Hence

$$\nabla \times F = \nabla \times (\nabla\varphi) = 0,$$

because the mixed second partial derivatives are equal. Therefore F is irrotational.
Now suppose that F is irrotational. Let $\Gamma \subset D$ be a closed curve. Since D is simply connected, Γ bounds a surface $\Sigma \subset D$. By Stokes's formula,

$$\oint_\Gamma F \cdot d\mathbf{p} = \iint_\Sigma (\operatorname{rot} F) \cdot \mathbf{n}\, d\sigma = 0.$$

Thus F is conservative.
Finally, suppose F is conservative. Fix

$$A = (a, b, c) \in D$$

and define

$$\varphi(x, y, z) = \int_A^{(x,y,z)} F \cdot d\mathbf{p}.$$

Since the integral is independent of path, φ is well-defined.
For sufficiently small h,

$$\begin{aligned}\varphi(x + h, y, z) - \varphi(x, y, z) \\ = \int_x^{x+h} P(t, y, z)\, dt.\end{aligned}$$

Hence, by the mean value theorem,

$$\frac{\varphi(x + h, y, z) - \varphi(x, y, z)}{h} = P(x^*, y, z)$$

for some x^* between x and $x + h$. Letting $h \to 0$,

$$\frac{\partial\varphi}{\partial x} = P(x, y, z).$$

Similarly,

$$\frac{\partial\varphi}{\partial y} = Q(x, y, z), \qquad \frac{\partial\varphi}{\partial z} = R(x, y, z).$$

Therefore

$$\nabla\varphi = F.$$

Thus F is a potential field. □

Brief summary. For a scalar field f,

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right),$$

and

$$\Delta f = \nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

For

$$F = (P, Q, R),$$

we have

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z},$$

and

$$\nabla \times F = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right).$$

1.5 Solenoidal Fields and Vector Potentials

Definition 1.5.1 (Solenoidal Field)**.** Let $D \subset \mathbb{R}^3$, and let F be a vector field on D. If there exists a vector field G such that

$$F = \nabla \times G,$$

then F is called a *solenoidal field*, and G is called a *vector potential* of F.

Theorem 1.5.2. Every solenoidal field is divergence-free:

$$F = \nabla \times G \quad \Longrightarrow \quad \nabla \cdot F = 0.$$

Proof. Let

$$G = (G_1, G_2, G_3).$$

Then

$$\nabla \times G = \left(\frac{\partial G_3}{\partial y} - \frac{\partial G_2}{\partial z}, \frac{\partial G_1}{\partial z} - \frac{\partial G_3}{\partial x}, \frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y} \right).$$

Therefore

$$\nabla \cdot (\nabla \times G) = 0$$

by equality of mixed partial derivatives. □

Thus

$$F \text{ solenoidal} \quad \Longrightarrow \quad F \text{ divergence-free.}$$

The converse is not true on an arbitrary domain.

Example 1.5.3. Let

$$D = \mathbb{R}^3 \setminus \{0\}, \qquad F(p) = \frac{p}{\|p\|^3}.$$

Then

$$\nabla \cdot F = 0$$

on D, but F is not a solenoidal field on all of D.
Indeed, let Σ_1 and Σ_2 be the upper and lower hemispheres of a sphere centered at the origin. Their boundaries are the same circle but have opposite orientations.
If $F = \nabla \times G$, then by Stokes's formula,

$$\iint_{\Sigma_1} F \cdot \mathbf{n} \, d\sigma = \oint_{\partial \Sigma_1} G \cdot d\mathbf{p},$$

while

$$\iint_{\Sigma_2} F \cdot \mathbf{n} \, d\sigma = - \oint_{\partial \Sigma_1} G \cdot d\mathbf{p}.$$

Thus the total flux through the sphere would be zero.
However,

$$\iint_{S_r^2} \frac{p}{\|p\|^3} \cdot \mathbf{n}\, d\sigma = \frac{1}{r^2} \iint_{S_r^2} d\sigma = 4\pi,$$

a contradiction.

Definition 1.5.4 (Star-Shaped Region)**.** Let $\Omega \subset \mathbb{R}^3$ and $p_0 \in \Omega$. If

$$\overline{p_0 p} \subset \Omega$$

for every $p \in \Omega$, then Ω is called *star-shaped with respect to* p_0.

Theorem 1.5.5. Let $D \subset \mathbb{R}^3$ be star-shaped with respect to p_0, and let $F \in C^1(D)$. Then

$$F \text{ is solenoidal} \quad \Longleftrightarrow \quad \nabla \cdot F = 0.$$

Sketch of Proof. The implication

$$F = \nabla \times G \quad \Longrightarrow \quad \nabla \cdot F = 0$$

has already been proved.
For the converse, translate the coordinate system so that $p_0 = 0$. Let

$$F = (P, Q, R).$$

Since D is star-shaped, if $(x, y, z) \in D$, then

$$(tx, ty, tz) \in D \qquad (0 \le t \le 1).$$

Define

$$G(x, y, z) = \int_0^1 tF(tx, ty, tz) \times (x, y, z)\, dt.$$

Equivalently,

$$G = (G_1, G_2, G_3),$$

where

$$G_1 = z \int_0^1 tQ(tx, ty, tz)\, dt - y \int_0^1 tR(tx, ty, tz)\, dt,$$

$$G_2 = x \int_0^1 tR(tx, ty, tz)\, dt - z \int_0^1 tP(tx, ty, tz)\, dt,$$

and

$$G_3 = y \int_0^1 tP(tx, ty, tz)\, dt - x \int_0^1 tQ(tx, ty, tz)\, dt.$$

Using

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0,$$

differentiate the components of G. For example,

$$\frac{\partial G_3}{\partial y} - \frac{\partial G_2}{\partial z} = P(x, y, z).$$

The remaining two components are obtained similarly. Hence

$$\nabla \times G = F.$$

□

If the center of the star-shaped region is

$$p_0 = (x_0, y_0, z_0),$$

one may write

$$G(x,y,z) = \int_0^1 tF\Big(x_0 + t(x-x_0), y_0 + t(y-y_0), \\ z_0 + t(z-z_0)\Big) \times (p - p_0)\, dt.$$

Proposition 1.5.6. If G is a vector potential of F, then for any differentiable scalar field φ,

$$G + \nabla\varphi$$

is also a vector potential of F.

Proof. Since

$$\nabla \times \nabla\varphi = 0,$$

we have

$$\nabla \times (G + \nabla\varphi) = \nabla \times G + \nabla \times \nabla\varphi = F.$$

□

1.6 Gradient, Divergence, and Curl in Orthogonal Curvilinear Coordinates

Consider a mapping

$$p = (x_1, x_2, x_3) = f(u_1, u_2, u_3)$$

from a domain D in the u-space into $\mathbb{R}^3$, where

$$x_i = x_i(u_1, u_2, u_3), \qquad i = 1, 2, 3.$$

Assume that:

1. $f \in C^1(D)$;
2. f is one-to-one;
3. the vectors

$$\frac{\partial f}{\partial u_1}, \qquad \frac{\partial f}{\partial u_2}, \qquad \frac{\partial f}{\partial u_3}$$

   are mutually orthogonal;

4. $$\det J_f > 0.$$

Then (u_1, u_2, u_3) forms an *orthogonal curvilinear coordinate system.*
For fixed u_2^0, u_3^0, the curve

$$p = f(u_1, u_2^0, u_3^0)$$

is called a u_1-coordinate curve. The u_2- and u_3-coordinate curves are defined similarly.
Define

$$h_i = \left\| \frac{\partial f}{\partial u_i} \right\|, \qquad \frac{\partial f}{\partial u_i} = h_i \mathbf{e}_i, \qquad i = 1, 2, 3,$$

where

$$\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$$

form an orthonormal frame.
The differential displacement is

$$d\mathbf{p} = \sum_{i=1}^{3} \frac{\partial f}{\partial u_i} du_i = \sum_{i=1}^{3} h_i \mathbf{e}_i \, du_i.$$

Theorem 1.6.1 (Gradient)**.** For a scalar field φ,

$$\nabla \varphi = \sum_{i=1}^{3} \frac{1}{h_i} \frac{\partial \varphi}{\partial u_i} \mathbf{e}_i.$$

Proof. Write

$$\nabla \varphi = \sum_{i=1}^{3} \lambda_i \mathbf{e}_i.$$

Since

$$d\varphi = \nabla \varphi \cdot d\mathbf{p},$$

we obtain

$$d\varphi = \sum_{i=1}^{3} \lambda_i h_i \, du_i.$$

On the other hand,

$$d\varphi = \sum_{i=1}^{3} \frac{\partial \varphi}{\partial u_i} du_i.$$

Hence

$$\lambda_i h_i = \frac{\partial \varphi}{\partial u_i},$$

which gives the formula. □

In particular,

$$\nabla u_i = \frac{\mathbf{e}_i}{h_i}, \qquad i = 1, 2, 3.$$

Since

$$\nabla \times \nabla u_i = 0,$$

we have

$$\nabla \times \left(\frac{\mathbf{e}_i}{h_i} \right) = 0.$$

Let

$$F = F_1 \mathbf{e}_1 + F_2 \mathbf{e}_2 + F_3 \mathbf{e}_3.$$

Theorem 1.6.2 (Divergence)**.** In orthogonal curvilinear coordinates,

$$\nabla \cdot F = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (F_1 h_2 h_3) + \frac{\partial}{\partial u_2} (F_2 h_3 h_1) + \frac{\partial}{\partial u_3} (F_3 h_1 h_2) \right].$$

Sketch of Proof. Write

$$F = F_1\mathbf{e}_1 + F_2\mathbf{e}_2 + F_3\mathbf{e}_3.$$

For the first component,

$$F_1\mathbf{e}_1 = F_1 h_2 h_3 \frac{\mathbf{e}_1}{h_2 h_3}.$$

Using the product rule for divergence and the identities of the orthogonal coordinate frame gives

$$\nabla \cdot (F_1\mathbf{e}_1) = \frac{1}{h_1 h_2 h_3} \frac{\partial (F_1 h_2 h_3)}{\partial u_1}.$$

The other two terms are obtained in the same way. □

Theorem 1.6.3 (Curl)**.** In orthogonal curvilinear coordinates,

$$\nabla \times F = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1\mathbf{e}_1 & h_2\mathbf{e}_2 & h_3\mathbf{e}_3 \\ \dfrac{\partial}{\partial u_1} & \dfrac{\partial}{\partial u_2} & \dfrac{\partial}{\partial u_3} \\ F_1 h_1 & F_2 h_2 & F_3 h_3 \end{vmatrix}.$$

Equivalently,

$$\begin{aligned} \nabla \times F = {} & \frac{1}{h_2 h_3} \left[\frac{\partial (F_3 h_3)}{\partial u_2} - \frac{\partial (F_2 h_2)}{\partial u_3} \right] \mathbf{e}_1 \\ & + \frac{1}{h_3 h_1} \left[\frac{\partial (F_1 h_1)}{\partial u_3} - \frac{\partial (F_3 h_3)}{\partial u_1} \right] \mathbf{e}_2 \\ & + \frac{1}{h_1 h_2} \left[\frac{\partial (F_2 h_2)}{\partial u_1} - \frac{\partial (F_1 h_1)}{\partial u_2} \right] \mathbf{e}_3. \end{aligned}$$

Finally, the Laplace operator is

$$\Delta \varphi = \nabla \cdot \nabla \varphi.$$

Hence, in orthogonal curvilinear coordinates,

$$\Delta \varphi = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^{3} \frac{\partial}{\partial u_i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial \varphi}{\partial u_i} \right).$$

Chapter 2

Numerical Series

2.1 Infinite Series and Their Convergence

Let

$$a_1, a_2, \ldots, a_n, \ldots$$

be a sequence of real numbers. The formal expression

$$a_1 + a_2 + \cdots + a_n + \cdots$$

is called an *infinite series*, denoted by

$$\sum_{n=1}^{\infty} a_n.$$

Definition 2.1.1 (Partial Sum)**.** The finite sum

$$S_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k$$

is called the n-th *partial sum* of the series.

Definition 2.1.2 (Convergence of a Series)**.** If the sequence of partial sums $\{S_n\}$ has a finite limit

$$\lim_{n\to\infty} S_n = S,$$

then the series

$$\sum_{n=1}^{\infty} a_n$$

is said to *converge*, and S is called its sum:

$$\sum_{n=1}^{\infty} a_n = S.$$

If $\{S_n\}$ does not have a finite limit, the series is said to *diverge*.

For $m > n$,

$$S_m - S_n = a_{n+1} + a_{n+2} + \cdots + a_m.$$

Therefore the convergence of a series is precisely the convergence of its sequence of partial sums.

Theorem 2.1.3 (Cauchy Criterion)**.** The series

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$m > n \geq N$$

implies

$$|a_{n+1} + a_{n+2} + \cdots + a_m| < \varepsilon.$$

Equivalently, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N, \qquad p \in \mathbb{N}$$

imply

$$|a_{n+1} + a_{n+2} + \cdots + a_{n+p}| < \varepsilon.$$

Proof. The series converges if and only if its partial sums $\{S_n\}$ form a convergent sequence. By the Cauchy criterion for sequences, this is equivalent to

$$|S_{n+p} - S_n| < \varepsilon$$

for all sufficiently large n, uniformly in $p \geq 1$. Since

$$S_{n+p} - S_n = a_{n+1} + \cdots + a_{n+p},$$

the result follows. □

Corollary 2.1.4 (Necessary Condition)**.** If

$$\sum_{n=1}^{\infty} a_n$$

converges, then

$$\lim_{n\to\infty} a_n = 0.$$

Indeed,

$$a_n = S_n - S_{n-1} \to S - S = 0.$$

Remark 2.1.5. The condition

$$a_n \to 0$$

is necessary but not sufficient for convergence.

For example,

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges although $1/n \to 0$.

Proposition 2.1.6. Adding, deleting, or changing finitely many terms of a series does not affect its convergence or divergence.

If

$$\sum_{n=1}^{\infty} a_n = A, \qquad \sum_{n=1}^{\infty} b_n = B,$$

then for $\alpha, \beta \in \mathbb{R}$,

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha A + \beta B.$$

2.2 Series with Positive Terms and the Comparison Tests

Definition 2.2.1 (Series with Positive Terms)**.** A series

$$\sum_{n=1}^{\infty} a_n$$

with

$$a_n \geq 0 \qquad (n = 1, 2, \ldots)$$

is called a *series with positive terms.*

For such a series,

$$S_{n+1} = S_n + a_{n+1} \geq S_n,$$

so the sequence of partial sums is monotonically increasing.

Theorem 2.2.2. A series with positive terms converges if and only if its sequence of partial sums is bounded above.

Proof. Since $\{S_n\}$ is increasing, this follows immediately from the monotone convergence theorem for sequences. □

Theorem 2.2.3 (Comparison Test)**.** Let

$$0 \leq a_n \leq b_n$$

for all sufficiently large n. Then:

1. if $\sum b_n$ converges, then $\sum a_n$ converges;
2. if $\sum a_n$ diverges, then $\sum b_n$ diverges.

Proof. For sufficiently large n, the corresponding tails satisfy

$$0 \leq \sum a_n \leq \sum b_n.$$

The conclusion follows from the boundedness criterion for positive series. □

Theorem 2.2.4 (Limit Comparison Test)**.** Let

$$a_n > 0, \qquad b_n > 0,$$

and suppose

$$\lim_{n\to\infty} \frac{a_n}{b_n} = l.$$

Then:

1. if $0 < l < \infty$, the two series
$$\sum a_n, \qquad \sum b_n$$
converge or diverge together;
2. if $l = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges;
3. if $l = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Sketch of Proof. If $0 < l < \infty$, choose positive constants c, C such that for all sufficiently large n,

$$c < \frac{a_n}{b_n} < C.$$

Thus

$$cb_n < a_n < Cb_n,$$

and the conclusion follows from the comparison test.
The cases $l = 0$ and $l = \infty$ are proved similarly. □

A basic comparison series is the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

Theorem 2.2.5. The p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if

$$p > 1.$$

2.3 Convergence Tests for Series with Positive Terms

Theorem 2.3.1 (Cauchy Root Test)**.** Let

$$a_n \geq 0, \qquad q = \limsup_{n\to\infty} \sqrt[n]{a_n}.$$

Then:

1. if $q < 1$, then $\sum a_n$ converges;
2. if $q > 1$, then $\sum a_n$ diverges;
3. if $q = 1$, the test gives no conclusion.

Proof. Suppose $q < 1$. Choose r such that

$$q < r < 1.$$

By the definition of the upper limit, for all sufficiently large n,

$$\sqrt[n]{a_n} < r,$$

hence

$$a_n < r^n.$$

Since the geometric series

$$\sum r^n$$

converges, so does $\sum a_n$.
If $q > 1$, there exist infinitely many n for which

$$\sqrt[n]{a_n} > 1.$$

Thus $a_n > 1$ for infinitely many n, and in particular $a_n \not\to 0$. Hence the series diverges. □

Theorem 2.3.2 (D'Alembert Ratio Test)**.** Let $a_n > 0$. If there exists $q < 1$ such that, for all sufficiently large n,

$$\frac{a_{n+1}}{a_n} \leq q,$$

then

$$\sum_{n=1}^{\infty} a_n$$

converges.
If, for all sufficiently large n,

$$\frac{a_{n+1}}{a_n} \geq 1,$$

then the series diverges.

A commonly used form is the following.

Corollary 2.3.3 (D'Alembert Ratio Test: Limit Form)**.** Suppose

$$a_n > 0$$

and

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = q.$$

Then:

$$q < 1 \quad \Longrightarrow \quad \sum a_n \text{ converges},$$

and

$$q > 1 \quad \Longrightarrow \quad \sum a_n \text{ diverges}.$$

When $q = 1$, the test gives no conclusion.

The same result remains valid in the form

$$\limsup_{n\to\infty} \frac{a_{n+1}}{a_n} < 1.$$

Theorem 2.3.4 (Raabe's Test)**.** Let $a_n > 0$. Suppose that, for all sufficiently large n,

$$n\left(\frac{a_n}{a_{n+1}} - 1\right) \geq r > 1.$$

Then

$$\sum a_n$$

converges.
If, for all sufficiently large n,

$$n\left(\frac{a_n}{a_{n+1}} - 1\right) \leq 1,$$

then

$$\sum a_n$$

diverges.

A convenient limit form is:

$$\lim_{n\to\infty} n\left(\frac{a_n}{a_{n+1}} - 1\right) = r.$$

Then

$$r > 1$$

implies convergence, whereas

$$r < 1$$

implies divergence.

Sketch of Proof. When $r > 1$, choose

$$1 < s < r.$$

For sufficiently large n,

$$\frac{a_n}{a_{n+1}} > 1 + \frac{s}{n}.$$

Comparison with a p-series of exponent $s > 1$ yields convergence.
The divergent case is obtained by comparison with the harmonic series. □

Theorem 2.3.5 (Gauss's Test)**.** Suppose $a_n > 0$, and

$$\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + O\left(\frac{1}{n^{1+\delta}}\right)$$

for some $\delta > 0$. Then:

1. if $\alpha > 1$, then $\sum a_n$ converges;
2. if $\alpha \leq 1$, then $\sum a_n$ diverges.

Sketch of Proof. The expansion implies

$$n\left(\frac{a_n}{a_{n+1}} - 1\right) = \alpha + o(1).$$

For $\alpha > 1$, Raabe's test gives convergence.
For $\alpha < 1$, Raabe's test gives divergence. The boundary case $\alpha = 1$ is handled by a more precise comparison with the harmonic series. □

2.4 Series with Terms of Arbitrary Sign

Let

$$\sum_{n=1}^{\infty} a_n$$

be a real series, where the terms are no longer assumed to be nonnegative.
The Cauchy criterion remains:

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N, \qquad p \in \mathbb{N}$$

imply

$$|a_{n+1} + \cdots + a_{n+p}| < \varepsilon.$$

Definition 2.4.1 (Alternating Series)**.** A series of the form

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n, \qquad a_n \geq 0,$$

is called an *alternating series*.

Theorem 2.4.2 (Leibniz Test)**.** Suppose

$$a_n \geq 0, \qquad a_{n+1} \leq a_n, \qquad a_n \to 0.$$

Then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

converges.

Proof. Let S_n denote the partial sums. Then

$$S_{2n} = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n}),$$

so

$$S_2 \leq S_4 \leq \cdots .$$

Also,

$$S_{2n+1} = a_1 - (a_2 - a_3) - \cdots - (a_{2n} - a_{2n+1}),$$

so

$$S_1 \geq S_3 \geq \cdots .$$

Moreover,

$$S_{2n+1} - S_{2n} = a_{2n+1} \to 0.$$

Hence the even and odd subsequences have the same limit, and the series converges. □

Lemma 2.4.3 (Abel Transformation)**.** Let

$$A_k = \sum_{j=1}^{k} a_j.$$

Then

$$\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n-1} A_k (b_k - b_{k+1}) + A_n b_n.$$

Proof. Since

$$a_k = A_k - A_{k-1}, \qquad A_0 = 0,$$

we have

$$\begin{aligned} \sum_{k=1}^{n} a_k b_k &= \sum_{k=1}^{n} (A_k - A_{k-1}) b_k \\ &= \sum_{k=1}^{n-1} A_k (b_k - b_{k+1}) + A_n b_n. \end{aligned}$$

□

Lemma 2.4.4 (Abel's Inequality)**.** Suppose

$$b_1 \geq b_2 \geq \cdots \geq b_n \geq 0$$

and

$$A_k = \sum_{j=1}^{k} a_j, \qquad |A_k| \leq M.$$

Then

$$\left| \sum_{k=1}^{n} a_k b_k \right| \leq M \left(|b_1| + 2|b_n| \right).$$

Proof. By Abel transformation,

$$\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n-1} A_k (b_k - b_{k+1}) + A_n b_n.$$

Hence

$$\begin{aligned}\left|\sum_{k=1}^{n} a_k b_k\right| &\leq M\sum_{k=1}^{n-1}(b_k - b_{k+1}) + M|b_n| \\ &= M(b_1 - b_n) + M|b_n| \\ &\leq M(|b_1| + 2|b_n|).\end{aligned}$$

□

Theorem 2.4.5 (Dirichlet Test)**.** Suppose:

1. the partial sums

$$A_n = \sum_{k=1}^{n} a_k$$

   are bounded;

2. b_n is monotone;

3.

$$b_n \to 0.$$

Then

$$\sum_{n=1}^{\infty} a_n b_n$$

converges.

Proof. Let

$$|A_n| \leq M.$$

For $m > n$, apply Abel's inequality to

$$a_{n+1}, \ldots, a_m.$$

The corresponding partial sums are bounded by $2M$. Hence

$$\left|\sum_{k=n+1}^{m} a_k b_k\right| \leq CM|b_{n+1}| \to 0.$$

The Cauchy criterion gives convergence. □

Theorem 2.4.6 (Abel Test)**.** Suppose:

1.

$$\sum_{n=1}^{\infty} a_n$$

   converges;

2. $\{b_n\}$ is monotone and bounded.

Then

$$\sum_{n=1}^{\infty} a_n b_n$$

converges.

Proof. Since $\{b_n\}$ is monotone and bounded, there exists

$$b = \lim_{n\to\infty} b_n.$$

Write

$$a_n b_n = a_n(b_n - b) + b a_n.$$

The series

$$\sum b a_n$$

converges. Moreover,

$$b_n - b \to 0$$

monotonically, so Dirichlet's test implies that

$$\sum a_n(b_n - b)$$

converges. Therefore $\sum a_n b_n$ converges. □

2.5 Absolute Convergence and Conditional Convergence

Definition 2.5.1 (Absolute Convergence)**.** The series

$$\sum_{n=1}^{\infty} a_n$$

is called *absolutely convergent* if

$$\sum_{n=1}^{\infty} |a_n|$$

converges.

Definition 2.5.2 (Conditional Convergence)**.** A convergent series

$$\sum_{n=1}^{\infty} a_n$$

is called *conditionally convergent* if

$$\sum_{n=1}^{\infty} |a_n|$$

diverges.

Theorem 2.5.3. Every absolutely convergent series is convergent.

Proof. For $m > n$,

$$\left| \sum_{k=n+1}^{m} a_k \right| \leq \sum_{k=n+1}^{m} |a_k|.$$

Since

$$\sum |a_n|$$

satisfies the Cauchy criterion, so does

$$\sum a_n.$$

□

Write

$$a_n^+ = \frac{|a_n| + a_n}{2}, \qquad a_n^- = \frac{|a_n| - a_n}{2}.$$

Then

$$a_n = a_n^+ - a_n^-,$$

and

$$|a_n| = a_n^+ + a_n^-,$$

where

$$a_n^+ \geq 0, \qquad a_n^- \geq 0.$$

Thus

$$\sum a_n = \sum a_n^+ - \sum a_n^-.$$

If $\sum a_n$ converges absolutely, then both

$$\sum a_n^+ \qquad \text{and} \qquad \sum a_n^-$$

converge.
If $\sum a_n$ converges conditionally, then

$$\sum a_n^+ = +\infty, \qquad \sum a_n^- = +\infty.$$

Theorem 2.5.4. Any rearrangement of an absolutely convergent series converges to the same sum.

Sketch of Proof. Let

$$\sum_{n=1}^{\infty} a_n = S$$

be absolutely convergent and let

$$a_{n_1}, a_{n_2}, \ldots$$

be a rearrangement.
Given $\varepsilon > 0$, choose N such that

$$\sum_{n>N} |a_n| < \varepsilon.$$

Once the rearranged partial sum contains the first N terms of the original series, its difference from S consists only of a subset of the tail. Therefore its absolute value is less than ε. □

Theorem 2.5.5 (Riemann Rearrangement Theorem)**.** Let

$$\sum_{n=1}^{\infty} a_n$$

be conditionally convergent. Then its terms can be rearranged so that the resulting series:

1. converges to any prescribed real number;
2. diverges to $+\infty$;
3. diverges to $-\infty$;
4. or oscillates without a finite or infinite limit.

Sketch of Proof. For a conditionally convergent series,

$$\sum a_n^+ = +\infty, \qquad \sum a_n^- = +\infty, \qquad a_n \to 0.$$

To obtain a prescribed value S, first add enough positive terms to make the partial sum exceed S, then enough negative terms to make it less than S, and repeat.
The overshoots tend to zero because $a_n \to 0$. Hence the resulting partial sums converge to S.
The other cases follow by choosing the positive and negative blocks differently. □

2.6 Multiplication of Series

Let

$$\sum_{n=0}^{\infty} a_n = A, \qquad \sum_{n=0}^{\infty} b_n = B.$$

Definition 2.6.1 (Cauchy Product)**.** Define

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

The series

$$\sum_{n=0}^{\infty} c_n$$

is called the *Cauchy product* of

$$\sum a_n \qquad \text{and} \qquad \sum b_n.$$

Thus

$$c_0 = a_0 b_0,$$

$$c_1 = a_0 b_1 + a_1 b_0,$$

and, in general,

$$c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0.$$

Theorem 2.6.2. If both

$$\sum a_n \qquad \text{and} \qquad \sum b_n$$

are series with nonnegative terms and converge to A and B, respectively, then their Cauchy product converges and

$$\sum_{n=0}^{\infty} c_n = AB.$$

Sketch of Proof. The rectangular partial sums satisfy

$$\left(\sum_{k=0}^{n} a_k\right)\left(\sum_{k=0}^{n} b_k\right) \to AB.$$

Since all terms are nonnegative, the triangular sums defining the partial sums of the Cauchy product are squeezed between appropriate rectangular sums. Hence their limit is also AB. □

Theorem 2.6.3 (Mertens's Theorem)**.** Suppose

$$\sum_{n=0}^{\infty} a_n = A$$

converges absolutely and

$$\sum_{n=0}^{\infty} b_n = B$$

converges. Then the Cauchy product

$$\sum_{n=0}^{\infty} c_n, \qquad c_n = \sum_{k=0}^{n} a_k b_{n-k},$$

converges and

$$\sum_{n=0}^{\infty} c_n = AB.$$

Proof. Set

$$A_n = \sum_{k=0}^{n} a_k, \qquad B_n = \sum_{k=0}^{n} b_k, \qquad C_n = \sum_{k=0}^{n} c_k.$$

Then

$$C_n = a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0.$$

Hence

$$C_n - AB = \sum_{k=0}^{n} a_k (B_{n-k} - B) - B \sum_{k=n+1}^{\infty} a_k.$$

Since $B_n \to B$, the sequence $\{B_n - B\}$ is bounded. Given $\varepsilon > 0$, choose N such that

$$\sum_{k>N} |a_k|$$

is sufficiently small. For the finitely many terms $k \leq N$,

$$B_{n-k} - B \to 0.$$

Therefore

$$C_n \to AB.$$

□

Theorem 2.6.4 (Abel's Theorem for Cauchy Products)**.** If

$$\sum a_n = A, \qquad \sum b_n = B,$$

and their Cauchy product

$$\sum c_n$$

converges, then

$$\sum c_n = AB.$$

2.7 Infinite Products

Consider an infinite product

$$\prod_{n=1}^{\infty} p_n = p_1 p_2 \cdots p_n \cdots .$$

Define its partial products by

$$P_n = \prod_{k=1}^{n} p_k.$$

Definition 2.7.1 (Convergence of an Infinite Product)**.** The infinite product

$$\prod_{n=1}^{\infty} p_n$$

is said to converge if

$$\lim_{n\to\infty} P_n = P$$

exists and

$$P \neq 0.$$

If

$$P_n \to 0,$$

the product is said to *diverge to zero.*
For an infinite product

$$\prod_{n=1}^{\infty} (1 + a_n),$$

a necessary condition for convergence is

$$a_n \to 0.$$

Indeed, if

$$P_n = P_{n-1}(1 + a_n)$$

and $P_n \to P \neq 0$, then

$$1 + a_n = \frac{P_n}{P_{n-1}} \to 1.$$

Assume from now on that

$$1 + a_n > 0$$

for all sufficiently large n. Then

$$\log P_n = \sum_{k=1}^{n} \log(1 + a_k).$$

Theorem 2.7.2. The infinite product

$$\prod_{n=1}^{\infty} (1 + a_n), \qquad 1 + a_n > 0,$$

converges if and only if the series

$$\sum_{n=1}^{\infty} \log(1 + a_n)$$

converges.

Proof. Since

$$P_n = \exp\left(\sum_{k=1}^{n} \log(1 + a_k)\right),$$

if

$$\sum_{n=1}^{\infty} \log(1 + a_n) = L \in \mathbb{R},$$

then

$$P_n \to e^L > 0.$$

Conversely, if

$$P_n \to P > 0,$$

then

$$\log P_n \to \log P,$$

so

$$\sum \log(1 + a_n)$$

converges. □

Since

$$\log(1 + x) \sim x \qquad (x \to 0),$$

we obtain the basic relation between infinite products and series.

Theorem 2.7.3. Suppose $a_n \to 0$ and $a_n \geq 0$ for all sufficiently large n. Then

$$\prod_{n=1}^{\infty} (1 + a_n)$$

converges if and only if

$$\sum_{n=1}^{\infty} a_n$$

converges.

Proof. Since

$$\lim_{x \to 0} \frac{\log(1 + x)}{x} = 1,$$

the positive series

$$\sum a_n \qquad \text{and} \qquad \sum \log(1 + a_n)$$

have the same convergence behavior. Apply the preceding theorem. □

If

$$-1 < a_n \leq 0$$

for all sufficiently large n, then similarly

$$\prod_{n=1}^{\infty} (1 + a_n)$$

converges if and only if

$$\sum_{n=1}^{\infty} a_n$$

converges.

Definition 2.7.4 (Absolute Convergence of an Infinite Product)**.** The infinite product

$$\prod_{n=1}^{\infty}(1+a_n)$$

is called *absolutely convergent* if

$$\prod_{n=1}^{\infty}(1+|a_n|)$$

converges.

Theorem 2.7.5. The infinite product

$$\prod_{n=1}^{\infty}(1+a_n)$$

converges absolutely if and only if

$$\sum_{n=1}^{\infty}|a_n|$$

converges.

Corollary 2.7.6. If

$$\sum_{n=1}^{\infty}|a_n|$$

converges and

$$a_n \neq -1,$$

then

$$\prod_{n=1}^{\infty}(1+a_n)$$

converges to a nonzero number.

Chapter 3

Sequences and Series of Functions

3.1 Sequences and Series of Functions

Let

$$u_1(x), u_2(x), \ldots, u_n(x), \ldots, \qquad x \in E,$$

be a sequence of functions defined on the same set E.

Definition 3.1.1 (Sequence of Functions)**.** The sequence

$$\{u_n(x)\}_{n=1}^{\infty}$$

is called a *sequence of functions* on E.

For each fixed $x \in E$,

$$u_1(x), u_2(x), \ldots, u_n(x), \ldots$$

is an ordinary numerical sequence.

Definition 3.1.2 (Pointwise Convergence)**.** Suppose that, for every $x \in E$, the numerical sequence $\{u_n(x)\}$ converges. Define

$$u(x) = \lim_{n\to\infty} u_n(x).$$

Then $\{u_n\}$ is said to *converge pointwise* to u on E.

In symbols,

$$u_n(x) \to u(x), \qquad x \in E.$$

The definition means that for every fixed $x \in E$ and every $\varepsilon > 0$, there exists

$$N = N(\varepsilon, x)$$

such that

$$n > N \quad \Longrightarrow \quad |u_n(x) - u(x)| < \varepsilon.$$

Remark 3.1.3. For pointwise convergence, the integer N may depend on both ε and the point x.

Example 3.1.4. On $E = [0, 1]$, let

$$u_n(x) = x^n.$$

Then

$$\lim_{n\to\infty} x^n = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

Thus the pointwise limit is

$$u(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

Now consider a formal series

$$u_1(x) + u_2(x) + \cdots + u_n(x) + \cdots .$$

Definition 3.1.5 (Series of Functions)**.** The expression

$$\sum_{n=1}^{\infty} u_n(x)$$

is called a *series of functions*.

Its n-th partial sum is

$$S_n(x) = \sum_{k=1}^{n} u_k(x).$$

Definition 3.1.6 (Convergence of a Series of Functions)**.** If the sequence of partial sums $\{S_n(x)\}$ converges pointwise on E, then

$$\sum_{n=1}^{\infty} u_n(x)$$

is said to converge on E. Its sum function is

$$S(x) = \lim_{n\to\infty} S_n(x).$$

3.2 Uniform Convergence

Pointwise convergence allows the index $N = N(\varepsilon, x)$ to depend on x. A stronger notion is obtained by requiring one N to work for every $x \in E$.

Definition 3.2.1 (Uniform Convergence)**.** A sequence of functions $\{u_n\}$ is said to *converge uniformly* to u on E if, for every $\varepsilon > 0$, there exists

$$N = N(\varepsilon)$$

such that

$$n > N$$

implies

$$|u_n(x) - u(x)| < \varepsilon$$

for every $x \in E$.
We write

$$u_n \rightrightarrows u \qquad \text{on } E.$$

Thus the essential difference is

$$\text{pointwise convergence: } N = N(\varepsilon, x),$$

whereas

$$\text{uniform convergence: } N = N(\varepsilon).$$

Theorem 3.2.2. The sequence $\{u_n\}$ converges uniformly to u on E if and only if

$$\lim_{n\to\infty} \sup_{x\in E} |u_n(x) - u(x)| = 0.$$

Proof. Set

$$B_n = \sup_{x \in E} |u_n(x) - u(x)|.$$

Uniform convergence means precisely that, for every $\varepsilon > 0$, eventually

$$|u_n(x) - u(x)| < \varepsilon$$

for every $x \in E$. This is equivalent to

$$B_n < \varepsilon$$

for all sufficiently large n, that is,

$$B_n \to 0.$$

□

Definition 3.2.3 (Uniform Convergence of a Series)**.** The series

$$\sum_{n=1}^{\infty} u_n(x)$$

is said to converge uniformly on E if its partial sums

$$S_n(x) = \sum_{k=1}^{n} u_k(x)$$

converge uniformly to its sum function $S(x)$.

Theorem 3.2.4 (Cauchy Criterion for Uniform Convergence)**.** The sequence $\{u_n\}$ converges uniformly on E if and only if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$m, n > N$$

implies

$$|u_n(x) - u_m(x)| < \varepsilon$$

for every $x \in E$.

Theorem 3.2.5 (Cauchy Criterion for a Series of Functions)**.** The series

$$\sum_{n=1}^{\infty} u_n(x)$$

converges uniformly on E if and only if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n > N, \qquad p \in \mathbb{N}$$

imply

$$|u_{n+1}(x) + \cdots + u_{n+p}(x)| < \varepsilon$$

for every $x \in E$.

An immediate necessary condition is

$$u_n(x) \rightrightarrows 0.$$

Theorem 3.2.6 (Weierstrass Test)**.** Suppose

$$|u_n(x)| \le a_n \qquad (x \in E,\ n = 1, 2, \ldots),$$

where $a_n \ge 0$ and the numerical series

$$\sum_{n=1}^{\infty} a_n$$

converges. Then

$$\sum_{n=1}^{\infty} u_n(x)$$

converges uniformly and absolutely on E.

Proof. For $n > N$,

$$\begin{aligned}\left|\sum_{k=n+1}^{n+p} u_k(x)\right| &\le \sum_{k=n+1}^{n+p} |u_k(x)| \\ &\le \sum_{k=n+1}^{n+p} a_k.\end{aligned}$$

Since $\sum a_n$ satisfies the Cauchy criterion, the last quantity is smaller than ε for all sufficiently large n, independently of x. Hence the function series converges uniformly. □

Theorem 3.2.7 (Dirichlet Test)**.** Suppose

$$\sum_{n=1}^{\infty} a_n(x) b_n(x)$$

is a series of functions on E, and assume that:

1. the partial sums
$$A_n(x) = \sum_{k=1}^{n} a_k(x)$$
are uniformly bounded on E, that is,
$$|A_n(x)| \le M$$
for every $x \in E$ and $n \in \mathbb{N}$;
2. for each $x \in E$,
$$b_n(x)$$
is monotone in n;
3. $$b_n(x) \rightrightarrows 0$$
on E.

Then

$$\sum_{n=1}^{\infty} a_n(x) b_n(x)$$

converges uniformly on E.

Sketch of Proof. For a tail of the series, Abel's transformation gives

$$\sum_{k=n}^{m} a_k(x)b_k(x).$$

The partial sums of $a_k(x)$ are uniformly bounded by a constant depending only on M. Abel's inequality therefore gives

$$\left|\sum_{k=n}^{m} a_k(x)b_k(x)\right| \leq 2M\left(|b_n(x)| + 2|b_m(x)|\right).$$

Since $b_n \rightrightarrows 0$, the right-hand side tends uniformly to zero. The Cauchy criterion proves uniform convergence. □

Theorem 3.2.8 (Abel Test)**.** Suppose:

1. $$\sum_{n=1}^{\infty} a_n(x)$$
   converges uniformly on E;
2. $b_n(x)$ is monotone in n for every $x \in E$;
3. there exists $M > 0$ such that
   $$|b_n(x)| \leq M$$
   for every $x \in E$ and $n \in \mathbb{N}$.

Then

$$\sum_{n=1}^{\infty} a_n(x)b_n(x)$$

converges uniformly on E.

Sketch of Proof. By the uniform Cauchy criterion, the partial sums of every sufficiently late tail of

$$\sum a_n(x)$$

are uniformly small. Applying Abel's inequality to the corresponding tail of

$$\sum a_n(x)b_n(x)$$

gives a bound independent of x, proving uniform convergence. □

3.3 Properties of Uniformly Convergent Function Sequences and Series

The importance of uniform convergence is that it allows several properties of functions to pass to the limit.

Theorem 3.3.1 (Continuity of the Uniform Limit)**.** Suppose each f_n is continuous on E, and

$$f_n \rightrightarrows f$$

on E. Then f is continuous on E.

Proof. Fix $x_0 \in E$. For every $\varepsilon > 0$, choose n_0 such that

$$|f(x) - f_{n_0}(x)| < \frac{\varepsilon}{3}$$

for every $x \in E$.
Since f_{n_0} is continuous at x_0, there exists $\delta > 0$ such that

$$|x - x_0| < \delta$$

implies

$$|f_{n_0}(x) - f_{n_0}(x_0)| < \frac{\varepsilon}{3}.$$

Therefore

$$\begin{aligned} |f(x) - f(x_0)| &\leq |f(x) - f_{n_0}(x)| \\ &\quad + |f_{n_0}(x) - f_{n_0}(x_0)| \\ &\quad + |f_{n_0}(x_0) - f(x_0)| \\ &< \varepsilon. \end{aligned}$$

Hence f is continuous at x_0. □

Theorem 3.3.2 (Dini's Theorem)**.** Let

$$f_n \in C[a,b], \qquad f \in C[a,b].$$

Suppose that for every $x \in [a,b]$,

$$f_n(x) \to f(x),$$

and that the convergence is monotone in n. Then

$$f_n \rightrightarrows f$$

on $[a,b]$.

Sketch of Proof. Assume, without loss of generality, that

$$f_n(x) \uparrow f(x).$$

Set

$$g_n(x) = f(x) - f_n(x).$$

Then g_n is continuous,

$$g_n(x) \downarrow 0.$$

Given $\varepsilon > 0$, for every $x \in [a,b]$, choose N_x such that

$$g_{N_x}(x) < \varepsilon.$$

By continuity, the same inequality remains valid in some neighborhood of x.
These neighborhoods form an open cover of the compact interval $[a,b]$. Choose a finite subcover corresponding to $x_1, \ldots, x_m$, and put

$$N = \max\{N_{x_1}, \ldots, N_{x_m}\}.$$

By monotonicity,

$$0 \leq g_n(x) < \varepsilon$$

for every $x \in [a,b]$ and $n > N$. Thus convergence is uniform. □

Theorem 3.3.3 (Termwise Integration)**.** Suppose

$$f_n \in R[a,b],$$

and

$$f_n \rightrightarrows f$$

on $[a,b]$. Then $f \in R[a,b]$ and

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx.$$

Proof. For every $\varepsilon > 0$, choose N such that

$$|f_n(x) - f(x)| < \frac{\varepsilon}{b-a}$$

for every $x \in [a,b]$ whenever $n > N$. Then

$$\begin{aligned}\left|\int_a^b f_n(x)\,dx - \int_a^b f(x)\,dx\right| &\le \int_a^b |f_n(x) - f(x)|\,dx \\ &< \varepsilon.\end{aligned}$$

Hence

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx.$$

□

Consequently, if

$$\sum_{n=1}^{\infty} u_n(x)$$

converges uniformly on $[a,b]$, and every u_n is integrable, then

$$\int_a^b \sum_{n=1}^{\infty} u_n(x)\,dx = \sum_{n=1}^{\infty} \int_a^b u_n(x)\,dx.$$

Theorem 3.3.4 (Termwise Differentiation)**.** Suppose f_n is continuously differentiable on $[a,b]$, and:

1. there exists $x_0 \in [a,b]$ such that
$$\{f_n(x_0)\}$$
converges;
2. the sequence of derivatives
$$\{f_n'\}$$
converges uniformly on $[a,b]$ to a function g.

Then $\{f_n\}$ converges uniformly on $[a,b]$ to some function f, and

$$f'(x) = g(x).$$

Thus

$$\left(\lim_{n\to\infty} f_n(x)\right)' = \lim_{n\to\infty} f_n'(x).$$

Proof. Since $\{f_n(x_0)\}$ converges, it satisfies the Cauchy criterion. Since f_n' converges uniformly, for every $\varepsilon > 0$ there exists N such that, for $m, n > N$,

$$|f_m(x_0) - f_n(x_0)| < \frac{\varepsilon}{2}$$

and

$$|f_m'(x) - f_n'(x)| < \frac{\varepsilon}{2(b-a)}$$

for every $x \in [a, b]$.
By the Newton–Leibniz formula,

$$f_m(x) - f_n(x) = f_m(x_0) - f_n(x_0) + \int_{x_0}^{x} (f_m'(t) - f_n'(t))\, dt.$$

Hence

$$|f_m(x) - f_n(x)| < \varepsilon$$

uniformly in x. Thus $f_n \rightrightarrows f$.
Again,

$$f_n(x) = f_n(x_0) + \int_{x_0}^{x} f_n'(t)\, dt.$$

Passing to the limit gives

$$f(x) = f(x_0) + \int_{x_0}^{x} g(t)\, dt.$$

Therefore

$$f'(x) = g(x).$$

□

Consequently, if

$$\sum_{n=1}^{\infty} u_n(x)$$

converges at some point $x_0 \in [a, b]$, and

$$\sum_{n=1}^{\infty} u_n'(x)$$

converges uniformly on $[a, b]$, then

$$\left(\sum_{n=1}^{\infty} u_n(x)\right)' = \sum_{n=1}^{\infty} u_n'(x).$$

3.4 Power Series

A series of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n$$

is called a *power series* centered at x_0.
By replacing $x - x_0$ with x, we may first study

$$\sum_{n=0}^{\infty} a_n x^n.$$

Theorem 3.4.1 (Abel's First Theorem)**.** If the power series

$$\sum_{n=0}^{\infty} a_n x^n$$

converges at some point $x_0 \neq 0$, then it converges absolutely for every x satisfying

$$|x| < |x_0|.$$

Proof. Since

$$\sum_{n=0}^{\infty} a_n x_0^n$$

converges, the sequence

$$\{a_n x_0^n\}$$

is bounded. Thus there exists $M > 0$ such that

$$|a_n x_0^n| \leq M.$$

For $|x| < |x_0|$,

$$|a_n x^n| = |a_n x_0^n| \left|\frac{x}{x_0}\right|^n \leq M \left|\frac{x}{x_0}\right|^n.$$

The geometric series on the right converges. Therefore

$$\sum a_n x^n$$

converges absolutely. □

Corollary 3.4.2. If

$$\sum a_n x^n$$

diverges at x_1, then it diverges at every x satisfying

$$|x| > |x_1|.$$

Hence there exists a number

$$R \in [0, \infty]$$

such that:

$$|x| < R \quad \Longrightarrow \quad \sum a_n x^n \text{ converges absolutely,}$$

and

$$|x| > R \quad \Longrightarrow \quad \sum a_n x^n \text{ diverges.}$$

Definition 3.4.3 (Radius and Interval of Convergence)**.** The number R is called the *radius of convergence* of the power series.
The interval

$$(-R, R)$$

is called its *interval of convergence*.

The endpoints $x = \pm R$ must be considered separately.

Theorem 3.4.4 (Cauchy–Hadamard Formula)**.** Let

$$\lambda = \limsup_{n\to\infty} \sqrt[n]{|a_n|}.$$

Then the radius of convergence is

$$R = \frac{1}{\lambda},$$

with the conventions

$$\frac{1}{0} = +\infty, \qquad \frac{1}{+\infty} = 0.$$

Proof. For the series

$$\sum |a_n x^n|,$$

the Cauchy root test gives

$$\limsup_{n\to\infty} \sqrt[n]{|a_n x^n|} = |x| \limsup_{n\to\infty} \sqrt[n]{|a_n|} = |x|\lambda.$$

Hence the series converges if

$$|x|\lambda < 1$$

and diverges if

$$|x|\lambda > 1.$$

Therefore

$$R = \frac{1}{\lambda}.$$

□

If the limit exists, the ratio test gives the useful formula

$$R = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right|.$$

Theorem 3.4.5. Let

$$\sum_{n=0}^{\infty} a_n x^n$$

have radius of convergence $R > 0$. Then for every

$$0 < r < R,$$

the series converges uniformly on

$$[-r, r].$$

Proof. Choose x_0 such that

$$r < |x_0| < R.$$

Since

$$\sum a_n x_0^n$$

converges, there exists $M > 0$ such that

$$|a_n x_0^n| \le M.$$

For $|x| \le r$,

$$|a_n x^n| \le M \left(\frac{r}{|x_0|} \right)^n.$$

The geometric series on the right converges, so the Weierstrass test gives uniform convergence. □

Thus the sum

$$S(x) = \sum_{n=0}^{\infty} a_n x^n$$

is continuous on $(-R, R)$.

Theorem 3.4.6. If

$$S(x) = \sum_{n=0}^{\infty} a_n x^n$$

has radius of convergence R, then inside $(-R, R)$,

$$S'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

The differentiated power series has the same radius of convergence R.

More generally,

$$S^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1) a_n x^{n-k}.$$

Sketch of Proof. The differentiated series has coefficients na_n. Since

$$\sqrt[n]{n} \to 1,$$

the Cauchy–Hadamard formula shows that its radius of convergence is still R.
On every compact interval

$$[-r, r] \subset (-R, R),$$

the differentiated series converges uniformly. Therefore termwise differentiation is valid. □

Similarly,

$$\int_0^x S(t)\, dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}, \qquad |x| < R.$$

Theorem 3.4.7 (Abel's Second Theorem)**.** Suppose

$$\sum_{n=0}^{\infty} a_n R^n$$

converges for some $R > 0$. Then the sum function

$$S(x) = \sum_{n=0}^{\infty} a_n x^n$$

is continuous from the left at R:

$$\lim_{x \to R^-} S(x) = \sum_{n=0}^{\infty} a_n R^n.$$

Equivalently, after the substitution $x = Rt$,

$$\lim_{t \to 1^-} \sum_{n=0}^{\infty} a_n R^n t^n = \sum_{n=0}^{\infty} a_n R^n.$$

Theorem 3.4.8 (Uniqueness of Power-Series Representation)**.** Suppose

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} b_n x^n$$

for every x in a neighborhood of 0. Then

$$a_n = b_n$$

for every $n \geq 0$.

Indeed, evaluating successive derivatives at $x = 0$ gives

$$a_n = \frac{S^{(n)}(0)}{n!} = b_n.$$

3.5 Expansion of Functions into Power Series

Suppose that in a neighborhood of x_0,

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

Since a power series may be differentiated term by term,

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1) a_n (x - x_0)^{n-k}.$$

Setting $x = x_0$ gives

$$f^{(k)}(x_0) = k! a_k.$$

Hence necessarily

$$a_k = \frac{f^{(k)}(x_0)}{k!}.$$

Therefore, if f can be expanded into a power series centered at x_0, that series must be

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n.$$

Definition 3.5.1 (Taylor Series)**.** The power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$$

is called the *Taylor series* of f at x_0.

When $x_0 = 0$, it becomes

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n,$$

which is called the *Maclaurin series* of f.
Taylor's formula gives

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x).$$

Therefore,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

holds if and only if

$$R_n(x) \to 0.$$

Theorem 3.5.2. Suppose there exist $R > 0$ and $M > 0$ such that

$$|f^{(n)}(x)| \leq M$$

for every $n \geq 0$ and every

$$x \in (x_0 - R, x_0 + R).$$

Then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

for

$$|x - x_0| < R.$$

Proof. Using the Lagrange form of the remainder,

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1},$$

where ξ lies between x_0 and x. Hence

$$|R_n(x)| \leq M\frac{|x - x_0|^{n+1}}{(n+1)!} \to 0.$$

Therefore the Taylor series converges to $f(x)$. □

The basic Maclaurin expansions include

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad x \in \mathbb{R},$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \qquad x \in \mathbb{R},$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \qquad x \in \mathbb{R},$$

$$\ln(1 + x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \qquad -1 < x \leq 1,$$

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \qquad |x| \leq 1,$$

and

$$(1 + x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n, \qquad |x| < 1,$$

where

$$\binom{\alpha}{n} = \frac{\alpha(\alpha - 1)\cdots(\alpha - n + 1)}{n!}.$$

Chapter 4

Improper Integrals

4.1 Improper Integrals on Unbounded Intervals

Let f be defined on $[a, +\infty)$, and suppose that $f \in R[a, A]$ for every $A > a$.

Definition 4.1.1 (Improper Integral on an Infinite Interval)**.** If the finite limit

$$\lim_{A \to +\infty} \int_a^A f(x)\, dx$$

exists, then the improper integral

$$\int_a^{+\infty} f(x)\, dx$$

is said to *converge*, and we define

$$\int_a^{+\infty} f(x)\, dx = \lim_{A \to +\infty} \int_a^A f(x)\, dx.$$

If the limit does not exist as a finite number, the improper integral is said to *diverge*.

Similarly,

$$\int_{-\infty}^b f(x)\, dx = \lim_{A \to -\infty} \int_A^b f(x)\, dx,$$

provided that the limit exists.
For an integral over the whole real line, choose any $c \in \mathbb{R}$.

Definition 4.1.2. The improper integral

$$\int_{-\infty}^{+\infty} f(x)\, dx$$

is said to converge if both

$$\int_{-\infty}^c f(x)\, dx$$

and

$$\int_c^{+\infty} f(x)\, dx$$

converge. In this case,

$$\int_{-\infty}^{+\infty} f(x)\, dx = \int_{-\infty}^c f(x)\, dx + \int_c^{+\infty} f(x)\, dx.$$

The convergence does not depend on the choice of c.

Proposition 4.1.3 (Linearity)**.** If

$$\int_a^{+\infty} f(x)\, dx \qquad \text{and} \qquad \int_a^{+\infty} g(x)\, dx$$

converge, then for any $\alpha, \beta \in \mathbb{R}$,

$$\int_a^{+\infty} (\alpha f(x) + \beta g(x))\, dx$$

also converges and

$$\int_a^{+\infty} (\alpha f(x) + \beta g(x))\, dx = \alpha \int_a^{+\infty} f(x)\, dx + \beta \int_a^{+\infty} g(x)\, dx.$$

Proposition 4.1.4. If

$$0 \leq f(x) \leq g(x)$$

for all sufficiently large x, then:

1. if
$$\int_a^{+\infty} g(x)\,dx$$
converges, then
$$\int_a^{+\infty} f(x)\,dx$$
converges;
2. if
$$\int_a^{+\infty} f(x)\,dx$$
diverges, then
$$\int_a^{+\infty} g(x)\,dx$$
diverges.

Theorem 4.1.5 (Comparison by Limit)**.** Suppose

$$f(x) > 0, \qquad g(x) > 0$$

for all sufficiently large x, and

$$\lim_{x\to+\infty} \frac{f(x)}{g(x)} = l.$$

Then:

1. if $0 < l < \infty$, the integrals
$$\int_a^{+\infty} f(x)\,dx, \qquad \int_a^{+\infty} g(x)\,dx$$
converge or diverge together;
2. if $l = 0$ and
$$\int_a^{+\infty} g(x)\,dx$$
converges, then
$$\int_a^{+\infty} f(x)\,dx$$
converges;
3. if $l = +\infty$ and
$$\int_a^{+\infty} g(x)\,dx$$
diverges, then
$$\int_a^{+\infty} f(x)\,dx$$
diverges.

A basic comparison integral is

$$\int_a^{+\infty} \frac{dx}{x^p}, \qquad a > 0.$$

Theorem 4.1.6. The integral

$$\int_a^{+\infty} \frac{dx}{x^p}$$

converges if and only if

$$p > 1.$$

Proof. For $p \neq 1$,

$$\int_a^A x^{-p}\,dx = \frac{A^{1-p} - a^{1-p}}{1-p}.$$

If $p > 1$, then

$$A^{1-p} \to 0,$$

so the integral converges.
If $p < 1$, then

$$A^{1-p} \to +\infty,$$

and the integral diverges.
For $p = 1$,

$$\int_a^A \frac{dx}{x} = \log A - \log a \to +\infty.$$

□

4.2 Cauchy, Dirichlet, and Abel Tests

Theorem 4.2.1 (Cauchy Criterion)**.** The improper integral

$$\int_a^{+\infty} f(x)\,dx$$

converges if and only if, for every $\varepsilon > 0$, there exists $A_0 > a$ such that

$$A'', A' > A_0$$

implies

$$\left| \int_{A'}^{A''} f(x)\,dx \right| < \varepsilon.$$

Proof. Set

$$F(A) = \int_a^A f(x)\,dx.$$

Then

$$\int_a^{+\infty} f(x)\,dx$$

converges if and only if $F(A)$ has a finite limit as $A \to +\infty$. By the Cauchy criterion for functions, this is equivalent to

$$|F(A'') - F(A')| < \varepsilon$$

for all sufficiently large A', A''. Since

$$F(A'') - F(A') = \int_{A'}^{A''} f(x)\,dx,$$

the result follows. □

Corollary 4.2.2. If

$$\int_a^{+\infty} |f(x)|\,dx$$

converges, then

$$\int_a^{+\infty} f(x)\,dx$$

converges.

Indeed,

$$\left|\int_{A'}^{A''} f(x)\,dx\right| \leq \int_{A'}^{A''} |f(x)|\,dx.$$

Definition 4.2.3 (Absolute Convergence)**.** The improper integral

$$\int_a^{+\infty} f(x)\,dx$$

is called *absolutely convergent* if

$$\int_a^{+\infty} |f(x)|\,dx$$

converges.

Definition 4.2.4 (Conditional Convergence)**.** If

$$\int_a^{+\infty} f(x)\,dx$$

converges but

$$\int_a^{+\infty} |f(x)|\,dx$$

diverges, then the integral is called *conditionally convergent*.

The following mean-value statements are useful in the convergence tests.

Theorem 4.2.5 (First Mean Value Theorem for Integrals)**.** Let $f \in R[a,b]$, and suppose g is monotone on $[a,b]$. If $g \geq 0$, then there exists

$$\xi \in [a,b]$$

such that

$$\int_a^b f(x)g(x)\,dx = g(a)\int_a^{\xi} f(x)\,dx + g(b)\int_{\xi}^b f(x)\,dx.$$

A useful special case is: if g is monotone decreasing and $g \geq 0$, then there exists $\xi \in [a,b]$ such that

$$\int_a^b f(x)g(x)\,dx = g(a)\int_a^{\xi} f(x)\,dx.$$

Theorem 4.2.6. Let $f \in R[a,b]$, and suppose g is monotone on $[a,b]$. If

$$\left|\int_a^x f(t)\,dt\right| \leq M \qquad (a \leq x \leq b),$$

then

$$\left|\int_a^b f(x)g(x)\,dx\right| \leq M\left(|g(a)| + 2|g(b)|\right).$$

Sketch of Proof. Using the mean value theorem for integrals, write

$$\int_a^b f(x)g(x)\,dx = g(a)\int_a^\xi f(x)\,dx + g(b)\int_\xi^b f(x)\,dx$$

for some $\xi \in [a,b]$. Since

$$\left|\int_a^\xi f(x)\,dx\right| \le M$$

and

$$\left|\int_\xi^b f(x)\,dx\right| \le \left|\int_a^b f(x)\,dx\right| + \left|\int_a^\xi f(x)\,dx\right| \le 2M,$$

the desired estimate follows. □

Theorem 4.2.7 (Dirichlet Test)**.** Suppose:

1. $$F(A) = \int_a^A f(x)\,dx$$

   is bounded for $A \ge a$;

2. $g(x)$ is monotone;

3. $$\lim_{x\to+\infty} g(x) = 0.$$

Then

$$\int_a^{+\infty} f(x)g(x)\,dx$$

converges.

Proof. Suppose

$$|F(A)| \le M.$$

For $A'' > A' > a$, the partial integral of f on $[A', x]$ satisfies

$$\left|\int_{A'}^x f(t)\,dt\right| \le 2M.$$

By the preceding estimate,

$$\left|\int_{A'}^{A''} f(x)g(x)\,dx\right| \le 2M\left(|g(A')| + 2|g(A'')|\right).$$

Since $g(x) \to 0$, the right-hand side tends to zero as $A', A'' \to +\infty$. The Cauchy criterion gives convergence. □

Theorem 4.2.8 (Abel Test)**.** Suppose:

1. $$\int_a^{+\infty} f(x)\,dx$$

   converges;

2. $g(x)$ is monotone and bounded on $[a, +\infty)$.

Then

$$\int_a^{+\infty} f(x)g(x)\,dx$$

converges.

Proof. Since g is monotone and bounded,

$$\lim_{x\to+\infty} g(x) = l$$

exists. Write

$$g(x) = [g(x) - l] + l.$$

Then

$$g(x) - l \to 0$$

monotonically. By the Dirichlet test,

$$\int_a^{+\infty} f(x)[g(x) - l]\,dx$$

converges. Also,

$$l\int_a^{+\infty} f(x)\,dx$$

converges. Hence

$$\int_a^{+\infty} f(x)g(x)\,dx$$

converges. □

4.3 Improper Integrals with Singularities

Let f be defined on $(a, b]$, and suppose f is integrable on every interval $[a + \varepsilon, b]$, where $\varepsilon > 0$.

Definition 4.3.1 (Improper Integral at a Singular Endpoint)**.** If

$$\lim_{\varepsilon\to0^+} \int_{a+\varepsilon}^b f(x)\,dx$$

exists as a finite number, then

$$\int_a^b f(x)\,dx$$

is said to converge, and

$$\int_a^b f(x)\,dx = \lim_{\varepsilon\to0^+} \int_{a+\varepsilon}^b f(x)\,dx.$$

Similarly, if f is singular at b,

$$\int_a^b f(x)\,dx = \lim_{\varepsilon\to0^+} \int_a^{b-\varepsilon} f(x)\,dx.$$

If $c \in (a, b)$ is a singular point of f, then

$$\int_a^b f(x)\,dx$$

converges if and only if both

$$\int_a^c f(x)\,dx \qquad \text{and} \qquad \int_c^b f(x)\,dx$$

converge.

Theorem 4.3.2 (Cauchy Criterion)**.** Suppose f has a singularity at a. Then

$$\int_a^b f(x)\,dx$$

converges if and only if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < \delta' < \delta'' < \delta$$

implies

$$\left| \int_{a+\delta'}^{a+\delta''} f(x)\,dx \right| < \varepsilon.$$

The comparison tests, absolute convergence, Dirichlet test, and Abel test have corresponding forms for improper integrals with singularities.
A standard example is

$$\int_0^1 \frac{dx}{x^p}.$$

Theorem 4.3.3. The improper integral

$$\int_0^1 \frac{dx}{x^p}$$

converges if and only if

$$p < 1.$$

Two important improper integrals are the Beta and Gamma integrals.

Definition 4.3.4 (Beta Function)**.** For $p > 0$ and $q > 0$,

$$B(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,dx.$$

Definition 4.3.5 (Gamma Function)**.** For $s > 0$,

$$\Gamma(s) = \int_0^{+\infty} x^{s-1}e^{-x}\,dx.$$

Remark 4.3.6. The convergence of the Beta integral must be checked at both endpoints 0 and 1, while the Gamma integral must be checked both near 0 and as $x \to +\infty$.

Definition 4.3.7 (Cauchy Principal Value at Infinity)**.** If the limit

$$\lim_{A\to+\infty} \int_{-A}^{A} f(x)\,dx$$

exists, it is called the *Cauchy principal value* of

$$\int_{-\infty}^{+\infty} f(x)\,dx,$$

and is denoted by

$$\mathrm{P.V.} \int_{-\infty}^{+\infty} f(x)\,dx.$$

More generally,

$$\text{P. V.} \int_{-\infty}^{+\infty} f(x)\,dx = \lim_{A\to+\infty} \int_{-A}^{A} f(x)\,dx.$$

Definition 4.3.8 (Cauchy Principal Value at an Interior Singularity)**.** Suppose $c \in (a, b)$ is a singular point of f. If

$$\lim_{\varepsilon\to 0^+} \left[\int_a^{c-\varepsilon} f(x)\,dx + \int_{c+\varepsilon}^{b} f(x)\,dx \right]$$

exists, it is called the *Cauchy principal value* of the integral.

Remark 4.3.9. The existence of a Cauchy principal value does not imply that the ordinary improper integral converges.

4.4 Improper Multiple Integrals

Let

$$D \subset \mathbb{R}^2$$

be an unbounded region, and let

$$f : D \to \mathbb{R}$$

be continuous.
For $r > 0$, let

$$D_r = \left\{ (x, y) \in D : \sqrt{x^2 + y^2} \le r \right\}.$$

Definition 4.4.1 (Improper Double Integral)**.** If the finite limit

$$\lim_{r\to+\infty} \iint_{D_r} f(x, y)\,dx\,dy$$

exists, then the improper double integral of f over D is defined by

$$\iint_D f(x, y)\,dx\,dy = \lim_{r\to+\infty} \iint_{D_r} f(x, y)\,dx\,dy.$$

More generally, let

$$D_1 \subset D_2 \subset \cdots \subset D$$

be bounded regions such that

$$\bigcup_{n=1}^{\infty} D_n = D.$$

If the limit

$$\lim_{n\to\infty} \iint_{D_n} f(x, y)\,dx\,dy$$

exists and is independent of the particular exhaustion $\{D_n\}$, it defines the improper integral over D.
For nonnegative functions the situation is simpler.

Definition 4.4.2. Let

$$f(x, y) \ge 0$$

on D. Define

$$\iint_D f(x, y)\,dx\,dy = \sup_G \iint_G f(x, y)\,dx\,dy,$$

where the supremum is taken over all bounded measurable subregions $G \subset D$.

The improper integral is finite precisely when this supremum is finite.

Theorem 4.4.3 (Comparison Principle)**.** Suppose

$$0 \le f(x,y) \le g(x,y)$$

on D. If

$$\iint_D g(x,y)\,dx\,dy$$

converges, then

$$\iint_D f(x,y)\,dx\,dy$$

also converges.

Theorem 4.4.4. Suppose

$$|f(x,y)|$$

has a convergent improper integral over D. Then

$$\iint_D f(x,y)\,dx\,dy$$

also converges.

Proof. Since

$$-|f(x,y)| \le f(x,y) \le |f(x,y)|,$$

the result follows from the comparison principle applied to the positive and negative parts. □

Now suppose D is bounded, but f is unbounded near a point $p_0 \in D$. For $\delta > 0$, let

$$D_\delta = D \setminus B_\delta(p_0).$$

Definition 4.4.5 (Improper Integral near a Singular Point)**.** If

$$\lim_{\delta \to 0^+} \iint_{D_\delta} f(x,y)\,dx\,dy$$

exists, then

$$\iint_D f(x,y)\,dx\,dy$$

is defined to be this limit.

For a nonnegative function f, one may equivalently define

$$\iint_D f(x,y)\,dx\,dy = \sup_G \iint_G f(x,y)\,dx\,dy,$$

where G ranges over compact subregions on which f is bounded.

Remark 4.4.6. For sign-changing functions, absolute convergence is the natural condition ensuring that the value of the improper multiple integral does not depend on the way in which the boundary or singularity is approached.

Chapter 5

Fourier Analysis

5.1 Fourier Series and Fourier Coefficients

A periodic motion may be written in the form

$$x(t) = A\sin(nt + \varphi),$$

where A is the amplitude, $2\pi/n$ is the period, and φ is the initial phase.
The trigonometric system

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \ldots$$

is orthogonal on $[-\pi, \pi]$.
Indeed,

$$\int_{-\pi}^{\pi} \cos nx\, dx = 0, \qquad \int_{-\pi}^{\pi} \sin nx\, dx = 0,$$

and

$$\int_{-\pi}^{\pi} \cos mx \cos nx\, dx = \begin{cases} 0, & m \neq n, \\ \pi, & m = n \neq 0. \end{cases}$$

Similarly,

$$\int_{-\pi}^{\pi} \sin mx \sin nx\, dx = \begin{cases} 0, & m \neq n, \\ \pi, & m = n. \end{cases}$$

Also,

$$\int_{-\pi}^{\pi} \cos mx \sin nx\, dx = 0.$$

Definition 5.1.1 (Fourier Coefficients)**.** Let $f \in R[-\pi, \pi]$. Define

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\, dx,$$

and, for $n \geq 1$,

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\, dx,$$
$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\, dx.$$

These numbers are called the *Fourier coefficients* of f.

The corresponding Fourier series is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin nx\right).$$

We write formally

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin nx\right).$$

The symbol $\sim$ means that the coefficients on the right are the Fourier coefficients of f; it does not yet assert convergence of the series to f.

Theorem 5.1.2 (Riemann–Lebesgue Lemma)**.** If $f \in R[a, b]$, then

$$\lim_{\lambda\to\infty} \int_a^b f(x)\cos \lambda x\, dx = 0,$$

and

$$\lim_{\lambda\to\infty} \int_a^b f(x)\sin \lambda x\, dx = 0.$$

Corollary 5.1.3. If a_n, b_n are the Fourier coefficients of $f \in R[-\pi, \pi]$, then

$$a_n \to 0, \qquad b_n \to 0.$$

5.2 Convergence of Fourier Series

Let

$$S_n(x) = \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cos kx + b_k \sin kx)$$

be the n-th partial sum of the Fourier series of f.
Using the formulas for the Fourier coefficients,

$$\begin{aligned} S_n(x_0) = &\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx \\ &+ \frac{1}{\pi} \sum_{k=1}^{n} \int_{-\pi}^{\pi} f(x) \cos k(x - x_0)\, dx. \end{aligned}$$

Therefore

$$S_n(x_0) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) D_n(x - x_0)\, dx,$$

where

$$D_n(t) = \frac{1}{2} + \sum_{k=1}^{n} \cos kt$$

is called the *Dirichlet kernel*.

Lemma 5.2.1. The Dirichlet kernel satisfies

$$D_n(t) = \frac{\sin\left(n + \frac{1}{2}\right) t}{2 \sin(t/2)}.$$

Hence

$$S_n(x_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \frac{\sin\left(n + \frac{1}{2}\right)(x - x_0)}{\sin((x - x_0)/2)}\, dx.$$

Using periodicity and the substitution $t = x - x_0$,

$$S_n(x_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x_0 + t) \frac{\sin\left(n + \frac{1}{2}\right) t}{\sin(t/2)}\, dt.$$

Combining the positive and negative halves,

$$S_n(x_0) = \frac{1}{2\pi} \int_{0}^{\pi} \left[f(x_0 + t) + f(x_0 - t)\right] \frac{\sin\left(n + \frac{1}{2}\right) t}{\sin(t/2)}\, dt.$$

Since

$$\frac{1}{\pi} \int_{0}^{\pi} \frac{\sin\left(n + \frac{1}{2}\right) t}{2 \sin(t/2)}\, dt = \frac{1}{2},$$

the convergence problem is reduced to an oscillatory integral.

Theorem 5.2.2 (Dirichlet's Convergence Theorem)**.** Suppose f is 2π-periodic and piecewise smooth on $[-\pi, \pi]$. Then at every point x_0,

$$S_n(x_0) \to \frac{f(x_0 + 0) + f(x_0 - 0)}{2}.$$

In particular, if f is continuous at x_0, then

$$S_n(x_0) \to f(x_0).$$

Sketch of Proof. Set

$$S = \frac{f(x_0+0) + f(x_0-0)}{2}.$$

Then

$$S_n(x_0) - S = \frac{1}{2\pi}\int_0^\pi \frac{f(x_0+t) + f(x_0-t) - 2S}{\sin(t/2)} \cdot \sin\left(n+\frac{1}{2}\right)t\,dt.$$

If

$$\varphi(t) = \frac{f(x_0+t) + f(x_0-t) - 2S}{2\sin(t/2)},$$

then the assumptions ensure sufficient regularity of φ near 0. The Riemann–Lebesgue lemma gives

$$\int_0^\pi \varphi(t)\sin\left(n+\frac{1}{2}\right)t\,dt \to 0.$$

Hence

$$S_n(x_0) \to S.$$

□

A local version is useful.

Theorem 5.2.3 (Dini Criterion)**.** Suppose

$$S = \frac{f(x_0+0) + f(x_0-0)}{2},$$

and there exists $\delta > 0$ such that

$$\frac{f(x_0+t) + f(x_0-t) - 2S}{t}$$

is integrable on $(0, \delta]$. Then the Fourier series of f converges at x_0 to S.

Definition 5.2.4 (Lipschitz Condition)**.** The function f is said to satisfy a Lipschitz condition near x_0 if there exist $L > 0$, $\delta > 0$, and $0 < \alpha \le 1$ such that

$$|f(x_0+t) - f(x_0+0)| \le L|t|^\alpha$$

and

$$|f(x_0-t) - f(x_0-0)| \le L|t|^\alpha$$

for $0 < t < \delta$.

Theorem 5.2.5. If f satisfies a Lipschitz condition near x_0, then its Fourier series converges at x_0 to

$$\frac{f(x_0+0) + f(x_0-0)}{2}.$$

Corollary 5.2.6. If f is piecewise continuously differentiable on $[-\pi, \pi]$, then its Fourier series converges at every point x_0 to

$$\frac{f(x_0+0)+f(x_0-0)}{2}.$$

For a function on $[-l, l]$, the Fourier expansion becomes

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right],$$

where

$$a_n = \frac{1}{l} \int_{-l}^{l} f(x) \cos \frac{n\pi x}{l}\, dx,$$

and

$$b_n = \frac{1}{l} \int_{-l}^{l} f(x) \sin \frac{n\pi x}{l}\, dx.$$

5.3 Cesaro Summation and Fejer's Theorem

Let

$$S_0, S_1, \ldots, S_n, \ldots$$

be a numerical sequence. Define

$$\sigma_n = \frac{S_0 + S_1 + \cdots + S_n}{n+1}.$$

Definition 5.3.1 (Cesaro Summability)**.** If

$$\sigma_n \to S,$$

then the sequence $\{S_n\}$ is said to be *Cesaro summable* to S.

If $S_n \to S$ in the ordinary sense, then

$$\sigma_n \to S.$$

Thus ordinary convergence implies Cesaro convergence.
For the Fourier series of f, let

$$S_n(x)$$

be its partial sums and define

$$\sigma_n(x) = \frac{S_0(x) + S_1(x) + \cdots + S_n(x)}{n+1}.$$

Theorem 5.3.2 (Fejer's Theorem)**.** Let $f \in R[-\pi, \pi]$, and suppose that the one-sided limits $f(x_0+0)$ and $f(x_0-0)$ exist. Then

$$\sigma_n(x_0) \to \frac{f(x_0+0)+f(x_0-0)}{2}.$$

If f is continuous at x_0, then

$$\sigma_n(x_0) \to f(x_0).$$

Sketch of Proof. The Cesaro mean can be written as

$$\sigma_n(x_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x_0 + t) F_n(t)\, dt,$$

where

$$F_n(t) = \frac{1}{n+1} \left(\frac{\sin((n+1)t/2)}{\sin(t/2)} \right)^2$$

is the *Fejer kernel*.
It satisfies

$$F_n(t) \geq 0$$

and

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} F_n(t)\, dt = 1.$$

Let

$$S = \frac{f(x_0 + 0) + f(x_0 - 0)}{2}.$$

Then

$$\sigma_n(x_0) - S = \frac{1}{2\pi} \int_0^{\pi} \Big[f(x_0 + t) + f(x_0 - t) - 2S \Big] \cdot F_n(t)\, dt.$$

Near $t = 0$, the bracketed expression is small by the existence of the one-sided limits. Away from 0, the Fejer kernel tends uniformly to zero. Splitting the integral into these two parts gives

$$\sigma_n(x_0) \to S.$$

□

Corollary 5.3.3. If f is continuous and 2π-periodic, then its Fourier series is Cesaro summable to f uniformly on $\mathbb{R}$.

Theorem 5.3.4 (Weierstrass Approximation by Trigonometric Polynomials)**.** Every continuous 2π-periodic function can be uniformly approximated by trigonometric polynomials.

Indeed, the Fejer sums σ_n are trigonometric polynomials and satisfy

$$\sigma_n \rightrightarrows f.$$

5.4 L^2 Theory of Fourier Series

Let $f \in R[a, b]$. Introduce the inner product

$$(f, g) = \int_a^b f(x) g(x)\, dx$$

and the norm

$$\|f\|_2 = \left(\int_a^b |f(x)|^2\, dx \right)^{1/2}.$$

A system

$$\{\varphi_n\}$$

is called *orthonormal* if

$$(\varphi_m, \varphi_n) = \delta_{mn}.$$

Suppose

$$\varphi_1, \varphi_2, \ldots$$

is an orthonormal system. Define the Fourier coefficients

$$c_n = (f, \varphi_n) = \int_a^b f(x)\varphi_n(x)\,dx.$$

For

$$T_n = \sum_{k=1}^{n} a_k \varphi_k,$$

consider the quadratic error

$$\|f - T_n\|_2^2.$$

By orthogonality,

$$\begin{aligned}\|f - T_n\|_2^2 &= \|f\|_2^2 - 2\sum_{k=1}^{n} a_k c_k + \sum_{k=1}^{n} a_k^2 \\ &= \|f\|_2^2 - \sum_{k=1}^{n} c_k^2 + \sum_{k=1}^{n} (a_k - c_k)^2.\end{aligned}$$

Therefore the error is minimized when

$$a_k = c_k.$$

Theorem 5.4.1 (Best Approximation Property)**.** Among all linear combinations

$$T_n = \sum_{k=1}^{n} a_k \varphi_k,$$

the Fourier sum

$$S_n = \sum_{k=1}^{n} c_k \varphi_k$$

minimizes the L^2-error

$$\|f - T_n\|_2.$$

Theorem 5.4.2 (Bessel's Inequality)**.** For every orthonormal system $\{\varphi_n\}$,

$$\sum_{n=1}^{\infty} |(f, \varphi_n)|^2 \le \|f\|_2^2.$$

For the trigonometric system on $[-\pi, \pi]$, this becomes

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} \left(a_n^2 + b_n^2\right) \le \frac{1}{\pi} \int_{-\pi}^{\pi} |f(x)|^2\,dx.$$

Definition 5.4.3 (Complete Orthonormal System)**.** An orthonormal system $\{\varphi_n\}$ is called *complete* if

$$(f, \varphi_n) = 0 \qquad \text{for all } n$$

implies

$$f = 0$$

in the corresponding L^2-sense.

Theorem 5.4.4. The trigonometric system

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \ldots$$

is complete on $[-\pi, \pi]$.

Theorem 5.4.5 (Parseval's Identity)**.** For a complete orthonormal system,

$$\sum_{n=1}^{\infty} |(f, \varphi_n)|^2 = \|f\|_2^2.$$

For Fourier series on $[-\pi, \pi]$,

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} \left(a_n^2 + b_n^2\right) = \frac{1}{\pi} \int_{-\pi}^{\pi} |f(x)|^2 \, dx.$$

Corollary 5.4.6. If all Fourier coefficients of $f \in R[-\pi, \pi]$ vanish, then

$$f = 0$$

in the L^2-sense.

More generally, if $f, g \in R[-\pi, \pi]$ have Fourier coefficients

$$a_n, b_n$$

and

$$\alpha_n, \beta_n,$$

respectively, then

$$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) g(x) \, dx = \frac{a_0 \alpha_0}{2} + \sum_{n=1}^{\infty} \left(a_n \alpha_n + b_n \beta_n\right).$$

5.5 Fourier Integral and Fourier Transform

Let

$$f \in L^1(-\infty, +\infty).$$

For $u \geq 0$, define

$$a(u) = \frac{1}{\pi} \int_{-\infty}^{+\infty} f(t) \cos ut \, dt,$$

and

$$b(u) = \frac{1}{\pi} \int_{-\infty}^{+\infty} f(t) \sin ut \, dt.$$

The corresponding Fourier integral is

$$\int_0^{+\infty} [a(u) \cos ux + b(u) \sin ux] \, du.$$

Formally,

$$f(x) \sim \int_0^{+\infty} [a(u) \cos ux + b(u) \sin ux] \, du.$$

For $A > 0$, define

$$S(A, x) = \int_0^{A} [a(u) \cos ux + b(u) \sin ux] \, du.$$

Substituting the expressions for $a(u)$ and $b(u)$,

$$\begin{aligned} S(A,x) &= \frac{1}{\pi}\int_{-\infty}^{+\infty} f(t)\left[\int_0^A \cos u(t-x)\,du\right] dt \\ &= \frac{1}{\pi}\int_{-\infty}^{+\infty} f(t)\frac{\sin A(t-x)}{t-x}\,dt. \end{aligned}$$

After the substitution $t = x + s$,

$$S(A,x) = \frac{1}{\pi}\int_0^{+\infty} [f(x+s)+f(x-s)]\,\frac{\sin As}{s}\,ds.$$

Theorem 5.5.1 (Fourier Integral Theorem)**.** Suppose $f \in L^1(-\infty,+\infty)$, and the one-sided limits $f(x+0)$ and $f(x-0)$ exist. If a Dini-type condition holds at x, then

$$\lim_{A\to+\infty} S(A,x) = \frac{f(x+0)+f(x-0)}{2}.$$

In particular, at a point of continuity,

$$f(x) = \int_0^{+\infty} [a(u)\cos ux + b(u)\sin ux]\,du.$$

Thus

$$\begin{aligned} f(x) = \frac{1}{\pi}\int_0^{+\infty}\left[\int_{-\infty}^{+\infty} f(t)\cos ut\,dt\right]\cos ux\,du \\ + \frac{1}{\pi}\int_0^{+\infty}\left[\int_{-\infty}^{+\infty} f(t)\sin ut\,dt\right]\sin ux\,du. \end{aligned}$$

If f is even, then

$$b(u) = 0,$$

and

$$a(u) = \frac{2}{\pi}\int_0^{+\infty} f(t)\cos ut\,dt.$$

Hence

$$f(x) = \frac{2}{\pi}\int_0^{+\infty}\left[\int_0^{+\infty} f(t)\cos ut\,dt\right]\cos ux\,du.$$

This is the *Fourier cosine transform.*
If f is odd, then

$$a(u) = 0,$$

and

$$b(u) = \frac{2}{\pi}\int_0^{+\infty} f(t)\sin ut\,dt.$$

Hence

$$f(x) = \frac{2}{\pi}\int_0^{+\infty}\left[\int_0^{+\infty} f(t)\sin ut\,dt\right]\sin ux\,du.$$

This is the *Fourier sine transform.*
The Fourier integral may also be written in complex form.

Definition 5.5.2 (Fourier Transform)**.** Define

$$\widehat{f}(u) = \int_{-\infty}^{+\infty} f(t)e^{-iut}\,dt.$$

Then the inverse Fourier formula is

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \widehat{f}(u) e^{iux}\, du,$$

whenever the Fourier inversion theorem is applicable.
Thus

$$f \longmapsto \widehat{f}$$

is called the *Fourier transform*.
If

$$g(x) = f'(x)$$

and the boundary terms vanish at infinity, then integration by parts gives

$$\widehat{g}(u) = iu\, \widehat{f}(u).$$

More generally,

$$\widehat{f^{(n)}}(u) = (iu)^n \widehat{f}(u).$$

Definition 5.5.3 (Convolution)**.** For suitable functions f and g, define

$$(f * g)(x) = \int_{-\infty}^{+\infty} f(x-u) g(u)\, du.$$

The Fourier transform converts convolution into multiplication:

$$\widehat{f * g}(u) = \widehat{f}(u) \widehat{g}(u).$$

Under the corresponding normalization, multiplication in the physical variable becomes convolution in the frequency variable.

Chapter 6

Integrals Depending on a Parameter

6.1 Integrals Depending on a Parameter

Let

$$f : [a, b] \times [\alpha, \beta] \to \mathbb{R}$$

be continuous. For $u \in [\alpha, \beta]$, define

$$\Phi(u) = \int_a^b f(x, u)\, dx.$$

Then Φ is a function of the parameter u.

Theorem 6.1.1 (Continuity with Respect to a Parameter)**.** If

$$f \in C\Big([a, b] \times [\alpha, \beta]\Big),$$

then

$$\Phi(u) = \int_a^b f(x, u)\, dx$$

is continuous on $[\alpha, \beta]$.

Proof. Fix $u_0 \in [\alpha, \beta]$. Since f is continuous on the compact rectangle

$$[a, b] \times [\alpha, \beta],$$

it is uniformly continuous. Hence, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|u - u_0| < \delta$$

implies

$$|f(x, u) - f(x, u_0)| < \frac{\varepsilon}{b - a}$$

for every $x \in [a, b]$. Therefore

$$\begin{aligned} |\Phi(u) - \Phi(u_0)| &= \left| \int_a^b [f(x, u) - f(x, u_0)]\, dx \right| \\ &\leq \int_a^b |f(x, u) - f(x, u_0)|\, dx \\ &< \varepsilon. \end{aligned}$$

Thus Φ is continuous at u_0. □

Theorem 6.1.2 (Differentiation under the Integral Sign)**.** Suppose

$$f(x, u)$$

and

$$\frac{\partial f}{\partial u}(x, u)$$

are continuous on

$$[a, b] \times [\alpha, \beta].$$

Then

$$\Phi(u) = \int_a^b f(x, u)\, dx$$

is differentiable, and

$$\Phi'(u) = \int_a^b \frac{\partial f}{\partial u}(x, u)\, dx.$$

Proof. For $h \neq 0$,

$$\frac{\Phi(u+h)-\Phi(u)}{h} = \int_a^b \frac{f(x,u+h)-f(x,u)}{h}\,dx.$$

By the mean value theorem,

$$\frac{f(x,u+h)-f(x,u)}{h} = \frac{\partial f}{\partial u}(x,\xi),$$

where ξ lies between u and $u+h$.
Since $\partial f/\partial u$ is uniformly continuous on the compact rectangle, as $h \to 0$,

$$\frac{f(x,u+h)-f(x,u)}{h} \rightrightarrows \frac{\partial f}{\partial u}(x,u)$$

with respect to x. Therefore

$$\Phi'(u) = \int_a^b \frac{\partial f}{\partial u}(x,u)\,dx.$$

□

Repeated differentiation gives, under the corresponding continuity assumptions,

$$\Phi^{(n)}(u) = \int_a^b \frac{\partial^n f}{\partial u^n}(x,u)\,dx.$$

Theorem 6.1.3 (Interchange of the Order of Integration)**.** If

$$f \in C\Big([a,b]\times[\alpha,\beta]\Big),$$

then

$$\int_\alpha^\beta \left(\int_a^b f(x,u)\,dx\right) du = \int_a^b \left(\int_\alpha^\beta f(x,u)\,du\right) dx.$$

Equivalently,

$$\int_\alpha^\beta \int_a^b f(x,u)\,dx\,du = \int_a^b \int_\alpha^\beta f(x,u)\,du\,dx.$$

Now let

$$p, q \in C^1[\alpha,\beta],$$

with

$$a < p(u) < q(u) < b.$$

Define

$$\Phi(u) = \int_{p(u)}^{q(u)} f(x,u)\,dx.$$

Theorem 6.1.4 (Leibniz Rule)**.** Suppose f and f_u are continuous. Then

$$\begin{aligned}\Phi'(u) = &\int_{p(u)}^{q(u)} \frac{\partial f}{\partial u}(x,u)\,dx \\ &+ f(q(u),u)q'(u) - f(p(u),u)p'(u).\end{aligned}$$

Sketch of Proof. Write

$$\Phi(u) = \int_a^{q(u)} f(x,u)\,dx - \int_a^{p(u)} f(x,u)\,dx.$$

Differentiate each term by combining differentiation under the integral sign with the ordinary chain rule. This gives

$$\begin{aligned}\Phi'(u) = &\int_a^{q(u)} f_u(x,u)\,dx + f(q(u),u)q'(u)\\ &- \int_a^{p(u)} f_u(x,u)\,dx - f(p(u),u)p'(u),\end{aligned}$$

hence the stated formula. □

6.2 Uniform Convergence of Improper Integrals Depending on a Parameter

Let

$$f(x,u)$$

be defined on

$$[a,+\infty) \times [\alpha,\beta].$$

For every fixed u, consider

$$\int_a^{+\infty} f(x,u)\,dx.$$

For $A > a$, define

$$\Phi_A(u) = \int_a^A f(x,u)\,dx.$$

Definition 6.2.1 (Uniform Convergence)**.** The improper integral

$$\int_a^{+\infty} f(x,u)\,dx$$

is said to converge uniformly with respect to $u \in [\alpha,\beta]$ if there exists a function $\Phi(u)$ such that

$$\Phi_A(u) \rightrightarrows \Phi(u)$$

on $[\alpha,\beta]$ as $A \to +\infty$.
We then write

$$\Phi(u) = \int_a^{+\infty} f(x,u)\,dx.$$

In ε-language, this means that for every $\varepsilon > 0$, there exists $A_0 > a$, independent of u, such that

$$A > A_0$$

implies

$$\left|\int_A^{+\infty} f(x,u)\,dx\right| < \varepsilon$$

for every $u \in [\alpha,\beta]$.

Theorem 6.2.2 (Cauchy Criterion)**.** The improper integral

$$\int_a^{+\infty} f(x,u)\,dx$$

converges uniformly on $[\alpha,\beta]$ if and only if, for every $\varepsilon>0$, there exists $A_0>a$ such that for all

$$A'', A' > A_0$$

and every $u\in[\alpha,\beta]$,

$$\left|\int_{A'}^{A''} f(x,u)\,dx\right| < \varepsilon.$$

Theorem 6.2.3 (Weierstrass Test)**.** Suppose

$$|f(x,u)| \le F(x)$$

for every

$$x \ge a, \qquad u\in[\alpha,\beta],$$

and suppose

$$\int_a^{+\infty} F(x)\,dx$$

converges. Then

$$\int_a^{+\infty} f(x,u)\,dx$$

converges uniformly and absolutely on $[\alpha,\beta]$.

Proof. For $A''>A'>A_0$,

$$\begin{aligned}\left|\int_{A'}^{A''} f(x,u)\,dx\right| &\le \int_{A'}^{A''} |f(x,u)|\,dx\\ &\le \int_{A'}^{A''} F(x)\,dx.\end{aligned}$$

The last quantity can be made smaller than any prescribed $\varepsilon>0$, independently of u. Hence the uniform Cauchy criterion applies. □

Theorem 6.2.4 (Dirichlet Test)**.** Suppose:

1. the integrals
$$\int_a^{A} f(x,u)\,dx$$
are uniformly bounded with respect to $A\ge a$ and $u\in[\alpha,\beta]$, that is,
$$\left|\int_a^{A} f(x,u)\,dx\right| \le M;$$
2. for every u, $g(x,u)$ is monotone in x;
3. $$g(x,u) \rightrightarrows 0 \qquad (x\to+\infty)$$
uniformly for $u\in[\alpha,\beta]$.

Then

$$\int_a^{+\infty} f(x,u)g(x,u)\,dx$$

converges uniformly on $[\alpha,\beta]$.

Theorem 6.2.5 (Abel Test)**.** Suppose:

1. $$\int_a^{+\infty} f(x,u)\,dx$$
   converges uniformly on $[\alpha,\beta]$;
2. for every u, $g(x,u)$ is monotone in x;
3. $g(x,u)$ is uniformly bounded:
   $$|g(x,u)| \le M.$$

Then

$$\int_a^{+\infty} f(x,u)g(x,u)\,dx$$

converges uniformly on $[\alpha,\beta]$.

6.3 Properties of Uniformly Convergent Improper Integrals

Let

$$\Phi(u) = \int_a^{+\infty} f(x,u)\,dx.$$

Theorem 6.3.1 (Continuity)**.** Suppose $f(x,u)$ is continuous on

$$[a,+\infty)\times[\alpha,\beta],$$

and

$$\int_a^{+\infty} f(x,u)\,dx$$

converges uniformly on $[\alpha,\beta]$. Then

$$\Phi(u)$$

is continuous on $[\alpha,\beta]$.

Proof. For $A > a$, define

$$\Phi_A(u) = \int_a^{A} f(x,u)\,dx.$$

Each Φ_A is continuous. By uniform convergence,

$$\Phi_A \rightrightarrows \Phi.$$

The uniform limit of continuous functions is continuous. □

Theorem 6.3.2 (Interchange of Limit and Improper Integral)**.** Suppose

$$u \to u_0, \qquad f(x,u) \to g(x)$$

for every $x \ge a$, and suppose

$$\int_a^{+\infty} f(x,u)\,dx$$

converges uniformly for u near u_0. Then

$$\lim_{u\to u_0} \int_a^{+\infty} f(x,u)\,dx = \int_a^{+\infty} g(x)\,dx.$$

Theorem 6.3.3 (Integration with Respect to the Parameter)**.** Suppose

$$f(x,u)$$

is continuous on

$$[a,+\infty)\times[\alpha,\beta]$$

and

$$\int_a^{+\infty} f(x,u)\,dx$$

converges uniformly for $u\in[\alpha,\beta]$. Then

$$\int_\alpha^\beta \left(\int_a^{+\infty} f(x,u)\,dx\right)du = \int_a^{+\infty}\left(\int_\alpha^\beta f(x,u)\,du\right)dx.$$

Proof. For $A>a$,

$$\int_\alpha^\beta \left(\int_a^{A} f(x,u)\,dx\right)du = \int_a^{A}\left(\int_\alpha^\beta f(x,u)\,du\right)dx.$$

Let $A\to+\infty$. Uniform convergence permits passage to the limit under the outer integral, giving the result. □

There is also a version involving two improper integrations.

Theorem 6.3.4. Suppose f is continuous on

$$[a,+\infty)\times[\alpha,+\infty).$$

Assume that the relevant iterated improper integrals converge uniformly so that both sides are defined. Then

$$\int_\alpha^{+\infty} \left(\int_a^{+\infty} f(x,u)\,dx\right)du = \int_a^{+\infty}\left(\int_\alpha^{+\infty} f(x,u)\,du\right)dx.$$

For absolutely convergent integrals, the order of integration may also be interchanged.

Theorem 6.3.5. If

$$\int_\alpha^{+\infty}\int_a^{+\infty} |f(x,u)|\,dx\,du<+\infty,$$

then

$$\int_\alpha^{+\infty} \left(\int_a^{+\infty} f(x,u)\,dx\right)du = \int_a^{+\infty}\left(\int_\alpha^{+\infty} f(x,u)\,du\right)dx.$$

Theorem 6.3.6 (Differentiation under an Improper Integral)**.** Suppose:

1. $f(x,u)$ and $f_u(x,u)$ are continuous on

$$[a,+\infty)\times[\alpha,\beta];$$

2. for some $u_0\in[\alpha,\beta]$,

$$\int_a^{+\infty} f(x,u_0)\,dx$$

converges;

3.
$$\int_a^{+\infty} \frac{\partial f}{\partial u}(x,u)\,dx$$
converges uniformly on $[\alpha,\beta]$.

Then
$$\Phi(u) = \int_a^{+\infty} f(x,u)\,dx$$
is differentiable and
$$\Phi'(u) = \int_a^{+\infty} \frac{\partial f}{\partial u}(x,u)\,dx.$$

Sketch of Proof. For $A > a$, define
$$\Phi_A(u) = \int_a^{A} f(x,u)\,dx.$$
Then
$$\Phi_A'(u) = \int_a^{A} f_u(x,u)\,dx.$$
By assumption,
$$\Phi_A' \rightrightarrows \int_a^{+\infty} f_u(x,u)\,dx.$$
Since $\Phi_A(u_0)$ converges at one point u_0, the theorem on uniform convergence of derivatives implies that Φ_A converges to a differentiable function Φ, and
$$\Phi'(u) = \lim_{A\to+\infty} \Phi_A'(u).$$
Hence
$$\Phi'(u) = \int_a^{+\infty} f_u(x,u)\,dx.$$
□

6.4 Gamma and Beta Functions

Definition 6.4.1 (Gamma Function)**.** For $s > 0$, define
$$\Gamma(s) = \int_0^{+\infty} t^{s-1}e^{-t}\,dt.$$
The integral converges for every $s > 0$.
For positive integers n,
$$\Gamma(n) = (n-1)!.$$

Theorem 6.4.2 (Recurrence Formula)**.** For $s > 0$,
$$\Gamma(s+1) = s\Gamma(s).$$

Proof. By integration by parts,
$$\begin{aligned}\Gamma(s+1) &= \int_0^{+\infty} t^s e^{-t}\,dt \\ &= \left[-t^s e^{-t}\right]_0^{+\infty} + s\int_0^{+\infty} t^{s-1}e^{-t}\,dt \\ &= s\Gamma(s).\end{aligned}$$
□

Since

$$\Gamma(1) = 1,$$

we obtain

$$\Gamma(n) = (n-1)!.$$

Theorem 6.4.3. The Gamma function is log-convex on $(0, +\infty)$; that is, for $s_1, s_2 > 0$ and $0 < \lambda < 1$,

$$\Gamma(\lambda s_1 + (1-\lambda)s_2) \leq \Gamma(s_1)^{\lambda}\Gamma(s_2)^{1-\lambda}.$$

Proof. Write

$$\begin{aligned}\Gamma(\lambda s_1 + (1-\lambda)s_2) &= \int_0^{+\infty} t^{\lambda(s_1-1)+(1-\lambda)(s_2-1)}e^{-t}\,dt \\ &= \int_0^{+\infty} \left(t^{s_1-1}e^{-t}\right)^{\lambda}\left(t^{s_2-1}e^{-t}\right)^{1-\lambda}\,dt.\end{aligned}$$

Hölder's inequality gives the result. □

Theorem 6.4.4 (Bohr–Mollerup Characterization)**.** Suppose

$$f : (0, +\infty) \to (0, +\infty)$$

satisfies

$$\begin{aligned}f(1) &= 1, \\ f(x+1) &= xf(x),\end{aligned}$$

and f is log-convex. Then

$$f(x) = \Gamma(x).$$

Thus the Gamma function is uniquely determined by positivity, log-convexity, normalization at 1, and the recurrence relation.
A useful product representation is

$$\Gamma(x) = \lim_{n\to\infty} \frac{n!\,n^x}{x(x+1)\cdots(x+n)}.$$

Equivalently,

$$\frac{1}{\Gamma(x)} = xe^{\gamma x}\prod_{n=1}^{\infty}\left(1+\frac{x}{n}\right)e^{-x/n},$$

where γ is Euler's constant.

Definition 6.4.5 (Beta Function)**.** For $p > 0$ and $q > 0$, define

$$B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,dt.$$

Immediately,

$$B(p,q) = B(q,p).$$

Theorem 6.4.6 (Recurrence Relations for the Beta Function)**.** For $p, q > 0$,

$$B(p+1,q) = \frac{p}{p+q}B(p,q),$$

and

$$B(p,q+1) = \frac{q}{p+q}B(p,q).$$

Consequently,

$$B(p+1,q)+B(p,q+1)=B(p,q).$$

Theorem 6.4.7 (Beta–Gamma Relation)**.** For $p,q>0$,

$$B(p,q)=\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

Proof. Starting from

$$\Gamma(p)\Gamma(q)=\int_0^{+\infty}x^{p-1}e^{-x}\,dx\int_0^{+\infty}y^{q-1}e^{-y}\,dy,$$

write

$$\Gamma(p)\Gamma(q)=\int_0^{+\infty}\int_0^{+\infty}x^{p-1}y^{q-1}e^{-(x+y)}\,dx\,dy.$$

Use the substitution

$$x=rt,\qquad y=r(1-t),$$

where

$$r\in(0,+\infty),\qquad t\in(0,1).$$

The Jacobian is

$$\left|\frac{\partial(x,y)}{\partial(r,t)}\right|=r.$$

Hence

$$\begin{aligned}\Gamma(p)\Gamma(q)&=\int_0^{+\infty}r^{p+q-1}e^{-r}\,dr\int_0^1 t^{p-1}(1-t)^{q-1}\,dt\\&=\Gamma(p+q)B(p,q).\end{aligned}$$

Therefore

$$B(p,q)=\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

□

An equivalent form is

$$B(p,q)=\int_0^{+\infty}\frac{x^{p-1}}{(1+x)^{p+q}}\,dx.$$

Indeed, use

$$t=\frac{x}{1+x}.$$

Another useful form is obtained from

$$t=\sin^2\theta:$$

$$B(p,q)=2\int_0^{\pi/2}(\sin\theta)^{2p-1}(\cos\theta)^{2q-1}\,d\theta.$$

Taking $p=q=\frac{1}{2}$,

$$B\left(\frac{1}{2},\frac{1}{2}\right)=\pi.$$

Therefore

$$\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}.$$

Using the recurrence relation,

$$\Gamma\left(n+\frac{1}{2}\right)=\frac{(2n)!}{4^n n!}\sqrt{\pi}.$$

Equivalently,

$$\Gamma\left(n+\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi}.$$

The Beta function also gives the classical trigonometric integrals

$$\int_0^{\pi/2}\sin^m x\cos^n x\,dx=\frac{1}{2}B\left(\frac{m+1}{2},\frac{n+1}{2}\right).$$

In particular,

$$\int_0^{\pi/2}\sin^m x\,dx=\frac{1}{2}B\left(\frac{m+1}{2},\frac{1}{2}\right),$$

and

$$\int_0^{\pi/2}\cos^m x\,dx=\frac{1}{2}B\left(\frac{1}{2},\frac{m+1}{2}\right).$$

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